

Estimand Sensitivity in the Evaluation of Quantized Reasoning Checkpoints: Selection Bias, Seed Placebos, and Subset-Cost Instability

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3 October 2026

Abstract

Practitioners often choose a public quantized reasoning checkpoint from a single published number, but conclusions about such checkpoints depend on what is measured. We show where the estimand changes the answer—a single pass@1, a correctness-conditioned length estimate, and a single serving subset—using one pinned A100 stack (vLLM 0.7.0, eager [12]) and eight DeepSeek-R1-Distill checkpoints—Qwen-7B and Llama-8B in BF16, FP8 (executed as Marlin W8A16 [5]), community AWQ-4, and RedHatAI GPTQ-4—on MATH-500 [9, 15], GSM8K [2], and GPQA-Diamond [20] (88 runs, 56,408 completions).

(i) *Accuracy*. On MATH-500 and GSM8K only one drop survives item-level intervals and multiplicity control: Llama AWQ-4 (MATH-500 -2.76 pp; GSM8K -1.57 pp), and on MATH-500 it consists of finished wrong answers, not truncations or loops. Strict-majority voting over five samples (maj@5) lowers the point estimate of the MATH gap to -1.4 pp, a change that is not itself statistically established. (ii) *Length*. Under sampled decoding, the BF16-correct token-inflation contrast is biased by selection: comparing two BF16 seeds, with no quantization at all, already yields $+138$ (Qwen) and $+235$ (Llama) tokens, and conditioning on the quantized model’s success instead flips the point-estimate sign in four of six contrasts. (iii) *Cost*. A 20-prompt serving subset can reorder the cost ranking: random stratified subsets reproduce the campaign-length order in only 21% (Llama) and 40% (Qwen) of draws, and measured subset lengths exceed the campaign means on the same prompts for FP8 in all four family-by-condition groups (ratios 1.07–1.49), a discrepancy we cannot attribute; timing repeats that share one sampling seed cannot expose it. As a gold-free alternative to length heuristics, unanimous agreement of just two samples lowers selective risk to 0.7–2.4%, against 2.2–5.8% for a one-sample length rule at matched coverage; five samples reach at most 0.27%.

Keywords: Reasoning language models, quantization checkpoints, estimand sensitivity, selection effects, selective prediction.

1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Post-training quantization (FP8, AWQ-4 [16], GPTQ-4 [4]) is a common serving choice, and practitioners often pick a public checkpoint from a single published pass@1. Isolated pass@1 numbers hide whether the same checkpoint also changes trace length [14, 18, 19], termination, or the cost of a correct answer [3, 11]. Serving-stack details—engine version, decoding flags, CUDA graphs—can materially move reasoning scores [10]. This paper *pins* one stack rather than shifting it:

we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment. The unit of analysis is the public checkpoint $\times$ pinned serving stack $\times$ evaluation target.

We ask four questions under that pin:

- **RQ1.** Which pass@1 differences survive an item-level interval, and how much of a 4-bit gap does maj@5 remove?
- **RQ2.** Under sampled decoding, is a BF16-correct length increase a property of the checkpoint, or a selection effect of conditioning on quantized failures?
- **RQ3.** At matched coverage, does k -sample answer agreement have lower selective risk than a one-sample length rule, and how does that gain scale with k ?
- **RQ4.** How much of a subset serving-cost gap is one length draw that timing repeats do not cover?

Contributions. Each contribution answers the research question with the same number. (C1) **One robust accuracy drop.** Llama AWQ-4 is the only contrast that is Holm-significant on both MATH-500 and GSM8K; maj@5 lowers its point estimate by about half, a reduction that is not itself statistically established. AWQ results describe the tested `jakiAJK` artifacts, whose chat template, `float16` activations, and AWQ backend (resolved `quantization=awq`) are confounded with the format; none of these was isolated, so the results are not a statement about the AWQ method. The extra Llama AWQ-4 errors are finished wrong answers, not truncations or loops, whereas a small rise in truncations accounts for most of the (non-significant) Qwen 4-bit point gaps (Section 4.7). The Llama AWQ-4 drop is not detectable at difficulty levels 1–2 and is larger at levels 3–5. (C2) **A placebo-calibrated selection effect in the BF16-correct length estimator.** Under sampled decoding, a BF16-versus-BF16 seed placebo reproduces a large part of the BF16-correct token increase without any quantization, and the mirrored quant-correct conditional flips sign in four of six contrasts. (C3) **A cost curve for gold-free agreement.** Unanimity of $k = 2$ –5 samples has lower selective risk than a one-sample length rule at matched coverage at every k ; two samples already capture most of the reduction at twice the token cost. (C4) **Subset serving cost can be a length draw.** Random 20-prompt subsets reproduce the campaign-length cost order in only 21–40% of draws, because a few very long traces dominate a small subset mean; timing repeats share one sampling seed and cannot expose this. Campaign-length seconds, using total tokens over total time as tok/s, put Llama FP8 first on both conditions in all 10,000 item resamples; the 20-prompt Condition A order does not.

2 Related Work

GPTQ [4] and AWQ [16] were originally validated mainly on short-form benchmarks. Liu et al. (QRM; COLM 2025) [18] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization, reporting accuracy and mean length. Kurtic et al. [11] study vLLM serving efficiency across formats and batching regimes, and also report accuracy for DeepSeek-R1-distill models on AIME-24, MATH-500, and GPQA-Diamond (20 responses per query). Lian et al. [14] report token inflation as an unconditional ratio of mean chain-of-thought lengths (their CTIR is a ratio of mean chain-of-thought-only lengths under greedy decoding on 100 questions per benchmark; our ratio-of-means column counts whole completions at $T = 0.6$, so the two are related but not the same estimand) and, in a supplementary correctness-conditioned analysis, find inflation stronger on the BF16-correct subset; RQ2 examines that conditional. A seed placebo needs stochastic sampling and cannot be run under greedy decoding, so RQ2 concerns the sampled setting. Hochlehnert et al. [10] show that seeds, prompts, decoding, and software

Table 1: Qwen-7B numbers reported by QRM beside this pinned grid. QRM values are from its Table 1: arXiv v1 (April 2025, single values) and v2, the COLM 2025 version (August 2025, mean $\pm$ SD over seeds); AWQ is the W4 group-128 weight-only row. Our values are mean $\pm$ seed SD. QRM quantized its own AWQ-4 model, whereas this grid evaluates the public `jakiAJK` upload, so the AWQ rows compare different artifacts. The GPQA rows compare two different studies and are not paired contrasts; both used the same harness, so if QRM ran it with the harness’s default model seed its answer order may coincide with this study’s, which we did not verify. QRM’s own MATH-500 numbers moved between versions by more than the BF16–FP8 gaps in Table 4. The discrepancy is checkpoint- and stack-sensitive measurement, not a superiority claim.

Cell	QRM v1	QRM v2 (COLM)	This grid
MATH-500 BF16	94.60	93.9 $\pm$ 0.7	94.00 $\pm$ 0.55
MATH-500 AWQ-4	91.00	92.5 $\pm$ 1.4	93.12 $\pm$ 0.67
GPQA-Diamond BF16	50.00	51.0 $\pm$ 1.0	50.34 $\pm$ 2.96
GPQA-Diamond AWQ-4	50.00	48.1 $\pm$ 2.8	44.78 $\pm$ 3.04

move reasoning scores. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution.

Existing studies evaluate quantization accuracy, throughput, or individual reasoning behaviors. Kurtic et al. already show that format recommendations change between synchronous decode and continuous batching. Our Condition A versus Condition B contrast is a pinned-stack sensitivity on these public checkpoints, not a second batching law. Appendix Table 23 reports mean tokens divided by pass@1. Appendix Table 24 places this grid against those studies. The unit of comparison is the public checkpoint $\times$ pinned stack $\times$ evaluation target.

The FP8 and GPTQ-4 cells are public RedHatAI compressed-tensors checkpoints evaluated as A100 Marlin weight-only kernels. These are not Red Hat production-deployment measurements. We do not rerun SmoothQuant [22]. Lotfi et al. [19], Li et al. [13], Helcig et al. [8], Zhong et al. [23], and Ayadi et al. [1] study overthinking, reasoning degradation, statistically lossless quantization, calibration, and reliability scaling; we do not re-estimate those quantities. Self-consistency [21] and G-Pass@ k [17] are different estimands from the unique-mode rule in Section 4.8. We report mean tokens divided by pass@1, and an aggregate hybrid Cost-of-Pass proxy from measured GPU-seconds; neither is Erol et al.’s per-problem estimator.

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM’s Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128); **GPTQ-4** (group size 128). Hugging Face IDs and revisions are in Appendix Table 18. Table 2 records the activation dtype and kernel. An AWQ-versus-BF16 contrast is therefore quantization plus a change from `bfloat16` to `float16` activations. The two AWQ-4 uploads also ship a chat template whose generation prompt omits the `<think>\n` suffix that the other six templates append; Appendix Section A shows that the decoding context is identical once the model emits that marker itself, which it does in every AWQ-4 completion. *AWQ-4 backend and provenance.* The serving-confirmation engine logs record the resolved setting `quantization=awq` and the launcher passes no explicit override. In vLLM 0.7.0 the AWQ-Marlin conversion requires a runtime zero point, and both `jakiAJK` configurations set `zero_point=false`, so these checkpoints cannot be converted whatever the launch flag and the GEMM `AWQLinearMethod` is selected (no conversion

Table 2: Activation dtype and vLLM kernel. Accuracy dtype is the 56k campaign. Serving dtype is the confirmation timer. For AWQ-4 the backend is explained in Section 3.1. Prompt suffix (accuracy campaign): whether the chat template ends the generation prompt with `<think>\n` (Appendix Section A). GPTQ’s checkpoint file is compressed-tensors; the kernel is Marlin W4A16. Accuracy-campaign GPTQ uses bfloat16 and the timer uses float16. The serving timer passes stored subset prompts, which already contain the suffix, to every format.

Format	Accuracy dtype	Accuracy kernel	Serving dtype	Prompt suffix
BF16	bfloat16	native	bfloat16	in prompt
FP8	bfloat16	Marlin W8A16	bfloat16	in prompt
AWQ-4	float16	GEMM awq (resolved)	float16	model-emitted
GPTQ-4	bfloat16	Marlin W4A16	float16	in prompt

message appears in any log); this is a source-code inference consistent with the logs, and the kernel was not profiled. The same configurations carry `_name_or_path ./tmp_around_awq`, and the AutoAWQ default configuration uses zero points, so symmetric weights in AWQ format point to an AutoRound-style export rather than a default AutoAWQ run; the model cards document no calibration. We therefore read “AWQ-4” as a label for these two uploads, not for the AWQ algorithm. Qwen-family models have a known float16 overflow history; this grid does not include a BF16 checkpoint rerun at `dtype=float16`, so that mechanism is not identified. Accuracy-campaign GPTQ-4 stays on bfloat16. The serving script sets float16 for both AWQ-4 and GPTQ-4, so the GPTQ accuracy cells and the GPTQ timing cells do not share an activation dtype.

3.2 Benchmarks and protocol

MATH-500 [9, 15] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [2] ($n = 1,319$, seeds 42–44) and GPQA-Diamond [20] ($n = 198$, seeds 42–44): breadth. Decoding follows the QRM chat template in Appendix Section A, with $T = 0.6$, top- $p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), and max new tokens 32,768. Total: 56,408 completions in 88 checkpoint×benchmark×seed runs. Throughout, a *run* denotes one checkpoint×benchmark×seed evaluation; a *cell* denotes one checkpoint×benchmark aggregated over its evaluated seeds; a *format contrast* compares a quantized checkpoint with its matched BF16 checkpoint within the same model family and benchmark. MATH and GSM8K are widely used; a paired within-item design reduces but does not eliminate contamination interaction with format.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM `inference.py` path [18] with `gpu_memory_utilization=0.75`. Engine pin: `vLLM==0.7.0`, `enforce_eager=true`. Eager execution keeps the accuracy path reproducible and adds CPU kernel-launch overhead on single-stream decode. Production serving more often uses CUDA graphs. This paper has no CUDA-graph replicate, so Condition A format gaps are specific to eager execution. Effective launch settings are recorded in `results/reports/runtime_manifest.json`.

3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager. Primary numbers are the controlled confirmation (protocol: `docs/MEASURE_D_SERVING_CONFIRMATION_PROTOCOL.md`; raw JSON under `results/measured_serving_con`

firmation/raw/). Condition A is 20 sequential MATH-500 prompts (4 per difficulty level; both subsets are drawn from the campaign items, and the released subset files list each prompt with its `campaign_index`). Condition B is 100 prompts (20 per level) in one `llm.generate` call with `max_num_seqs=8`. The two conditions differ in both the prompt subset and the serving regime, so a ranking change between them is a property of the whole condition. With `max_num_seqs=8`, Condition B elapsed time runs until the last trace finishes. One or two long traces can decode alone after the batch drains. The raw JSON stores aggregate tokens and elapsed time, not time spent below a full batch of 8, so that tail share is not reported. A secondary microbenchmark forces 10 prompts to 512 tokens (`ignore_eos`). The timer passes the stored prompt strings to vLLM, which tokenizes them with a second beginning-of-sequence token: every request has one more input token than in the campaign (for example 1,986 versus 1,966 input tokens for the 20 Qwen Condition A prompts, in all four formats; the same holds in every cell and condition), so the AWQ-4 timer prompts differ from their own accuracy-campaign prompts by that token and by the `<think>\n` suffix that the AWQ-4 accuracy prompts lack.

Three warmup queries are discarded. Δt_{steady} is then the wall-clock from the first measured request to the return of the last measured response, bracketed by `torch.cuda.synchronize()`. Engine initialization is outside that interval. The first measured prefill and the straggler tail are inside it. Output tok/s = $(\sum \text{output tokens}) / \Delta t_{\text{steady}}$. GPU-sec/query = $\Delta t_{\text{steady}} / N$ on one GPU. Sampling matches the campaign ($T = 0.6$, top- $p = 0.95$) with seed 20260816. Repetitions share that seed, and the recorded aggregate output-token counts are identical across repeats (the generated sequences were not retained, so identity of the sequences is not verified). The tok/s spread is wall-clock jitter at a fixed aggregate length. It is not length variance. $R = 3$, expanded to $R = 5$ when the CV of the first three tok/s values exceeds 3%; as implemented, that CV uses the population standard deviation (Llama FP8 Condition A sits at 2.995% and keeps $R = 3$; the sample-SD CV would be 3.67% and would have triggered $R = 5$), whereas the reported tok/s spreads are sample SDs. With three repeats a percentile interval has at most ten distinct means. We therefore report the sample mean, the sample SD, and the observed min-max, and we do not treat those intervals as a stable estimate of timing noise. Qwen-7B jobs ran on `ragpu003` and Llama-8B jobs on `ragpu004`; both are two-GPU nodes, and each SLURM job requested one GPU and 16 CPU cores without `--exclusive`. The raw records log NVIDIA driver 550.54.14 on both hosts. Within a family, all four formats share a host but ran sequentially on one day (Qwen: BF16 05:36, FP8 08:29, AWQ-4 11:09, GPTQ-4 12:38 UTC), so format is also confounded with time on a node that other jobs could share. Eager decode is bound by kernel-launch CPU time, so host load moves single-stream tok/s. Across families, host and architecture are confounded. Archived logs do not contain per-run clocks, power, or co-tenant load. Peak VRAM is `torch.cuda.max_memory_allocated` after the 0.75 utilization preallocation.

$$\tilde{C}_{\text{pass}}^{\text{hyb}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \quad (1)$$

The numerator is subset GPU-seconds and the denominator is the frozen MATH-500 campaign pass@1. This ratio follows the Cost-of-Pass form and is not the per-problem estimator of Erol et al. [3]. Because the repeats share a seed, the printed interval multiplies wall-clock jitter by a pass@1 bootstrap. It does not resample prompt lengths. A second estimator holds measured tok/s fixed and substitutes the campaign mean length: campaign tokens $\div$ tok/s $\div$ pass@1, with tok/s taken as total tokens over total time across the repeats (the aggregate that matches the reported GPU-s/q and hybrid dollars; it differs from the mean of per-repeat ratios only for Qwen FP8 Condition B, which varied by 56% across repeats, and in the third digit elsewhere). That estimator still uses a length-dependent tok/s, so it is a sensitivity, not a replacement measurement. The \$1.50/A100-hour rate is a pricing scenario. Because it is a single constant, it rescales GPU-seconds without changing any ranking.

3.5 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is an item-level bootstrap of the paired difference $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$. For each problem i we first average correctness over seeds, $\delta_i = \overline{\text{acc}}_{i,\text{quant}} - \overline{\text{acc}}_{i,\text{BF16}}$, then resample n problems with replacement ($B = 10,000$; RNG seed 0). An independent reimplementation with a different random stream reproduced every contrast-interval endpoint to within about 0.1 pp on MATH-500 and GSM8K and 0.2 pp on GPQA-Diamond. Accuracy and stratum intervals use Python’s `random.Random(0)` stream; the placebo and per-level intervals use NumPy’s `default_rng(0)`. Seeds are averaged inside each item, then items are resampled. Resampling those item means carries the generation noise already averaged into them. The interval remains conditional on the seed set that was run. Appendix Table 25 adds a two-way bootstrap that also resamples the generation seeds. With bootstrap replicates Δ_b^* , the two-sided bootstrap tail-area p -value is

$$p_{\text{hi}} = \#\{\Delta_b^* \geq 0\}/B, \quad p_{\text{lo}} = \#\{\Delta_b^* \leq 0\}/B, \quad p = \min(1, 2 \min(p_{\text{hi}}, p_{\text{lo}})). \quad (2)$$

This is the p -value obtained by inverting the percentile interval. A reported $p = 0$ or $p < 0.001$ means that no sign-crossing bootstrap replicate was observed among the 10,000 draws; it is not a literal probability of zero. When a Holm decision lies within Monte Carlo error of α , we rerun that contrast with $B = 10^6$ and also report an item-level sign-flip permutation test (2×10^5 sign vectors) and a paired t -test on the item means. Holm–Bonferroni is applied separately to the six primary within-benchmark format contrasts (MATH-500, GSM8K, GPQA-Diamond). Each benchmark is a distinct inferential family because each corresponds to a separate task distribution; we do not pool the 18 contrasts for the primary analysis. A joint Holm correction over all 18 contrasts is reported only as a stricter sensitivity (Appendix Table 34). Two one-sided tests (TOST) at a ± 1 pp equivalence margin use the 90% CI; this is an equivalence check, not a default claim. We use ± 1 pp as a study tolerance, not a universal equivalence standard. The smallest margin that would contain the FP8 90% intervals is 1.12 pp for Qwen and 1.28 pp for Llama. A failed ± 1 pp test at this n reflects the sample size relative to that margin (see the minimum detectable effects in Section 4.1).

maj@5. A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold, i.e. if a strict majority of the five answers is correct. This is strict-majority voting; it is a lower bound on plurality voting, which can also be correct with two correct answers and three mutually different wrong ones. We report the accuracy, not only the McNemar test against BF16. Non-significant maj@5 tests do not imply that pass@1 effects are absent.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). The launcher sets both `max_model_len` and `max_new_tokens` to 32,768. In vLLM the first bound is prompt plus output, so the effective output cap is 32,768 minus that item’s prompt length. Compact JSON has no `finish_reason`. Tokenizing the stored campaign prompts with the frozen-runtime tokenizers (transformers 4.47.1) gives 29–811 tokens (Qwen) and 27–776 (Llama); the AWQ-4 prompts lack the two-token `<think>\n` suffix and are shorter (27–809 and 25–774). The effective cap therefore lies between 31,957 and 32,743 tokens. Re-tokenizing the stored completions with those tokenizers reproduces every stored `completion.tokens` value (56,408 of 56,408 completions); newer tokenizer releases do not (the Llama counts differ by about 220 tokens per completion), so the prompt-length and cap analyses must run in the frozen environment (`scripts/analysis/audit_checks.py`). We therefore report *near-cap* rows ($\geq 32,500$ tokens) as the truncation proxy. Because the stored prompts give each row its own cap, we also count *cap-band* rows whose completion lies within 16 tokens of that per-item cap (Section 4.3). The tolerance is wider than necessary: with the frozen tokenizers 147 of the 155 cap-band rows are exactly at the cap and 154 are within two tokens. The `boxed` flag is a string search for `\boxed`, which is separate from extractive match.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with an item-level percentile bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness, with item-level 95% CIs on each cell. A direct contrast $D = \bar{\Delta}_{\text{BF16-only}} - \bar{\Delta}_{\text{Both-OK}}$ uses the same item resamples for both stratum means; replicates in which either stratum is empty are skipped. D compares conditional means and is not a causal effect. A BF16-internal control compares BF16 length when BF16 is correct versus incorrect. The BF16-correct conditional Δ , following Lian et al., is the mean quantized-BF16 Δ on pairs where BF16 is correct ($\text{Both-OK} \cup \text{BF16-only}$). The mirrored quant-correct conditional is the same mean on pairs where the quantized checkpoint is correct ($\text{Both-OK} \cup \text{Quant-only}$). Conditioning on BF16-correct pulls in quantized failures, which are long, so the BF16-correct column is positive even when the unconditional change is near zero. Conditioning on quant-correct pulls in BF16 failures and moves the sign the other way. Both columns are selection effects. To size the selection effect directly we run a *seed placebo*: the same BF16-correct estimator applied to BF16 seed s against BF16 seed $t \neq s$, pooled over all 20 ordered seed pairs, with an item-level bootstrap CI. Any nonzero placebo value is produced by conditioning alone, because both arms are the same checkpoint. Let a_i be the BF16 accuracy on item i and μ_i^q, μ_i^b the mean lengths of the two checkpoints. If, given the item, the quantized run and the two BF16 seeds are independent, the BF16-correct conditional minus the placebo estimates $\sum_i a_i(\mu_i^q - \mu_i^b) / \sum_i a_i$, the unconditional change reweighted by BF16 accuracy; we report this reweighted change (with an item-level bootstrap CI) as a check. The same pooling gives the unconditional estimator a placebo of exactly zero, which is why the ratio of means and the unconditional paired Δ are the primary length estimands. D compares the BF16-only and Both-OK means on shared item resamples and is a mismatch diagnostic, not a causal token increase.

Tokens per correct answer. Appendix Table 23 reports campaign mean tokens divided by MATH-500 pass@1. That ratio is what a shared throughput would rank. It does not use a tok/s constant. Primary cost evidence is Table 15 together with the campaign-length sensitivity in Table 16. The hybrid dollar in Table 15 divides measured GPU-seconds by pass@1. For GPTQ-4 those GPU-seconds were timed at `float16`, while the pass@1 denominator comes from the `bfloat16` accuracy campaign. The two GPTQ cells are not the same configuration.

Observable modal agreement. Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement without gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). For item i and five generated predictions, let $\sim$ be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of $\sim$. Let m_i be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is $g_i = m_i/5$. Serve at threshold $k/5$ only when a unique modal class satisfies $m_i \geq k$. The pattern (A, A, B, B, C) has $m_i = 2$ and is not served at $\geq 3/5$; (A, A, A, B, B) has unique mode A with $m_i = 3$ and is served at $\geq 3/5$. Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy T_5 is the sum of their completion tokens, including abstained queries. We report three discrete operating points, not a continuous risk-coverage curve. Coverage and selective risk follow the selective-classification framework of Geifman and El-Yaniv [6]. For the cost curve we also use a k -sample unanimity rule for $k = 2, \dots, 5$: serve when all k answers fall in one equivalence class, and average coverage and risk over all $\binom{5}{k}$ seed subsets. Its token cost is k times the mean completion length.

Table 3: MATH-500 headline grid ($n = 500$, seeds 42–46). Item-level 95% CIs resample problems. maj@5: items with at least 3 of 5 seeds correct. Loops: identical-word run ≥ 20 . NC: near-cap, completion tokens $\geq 32,500$, the truncation proxy (Section 4.3).

Cell	42	43	44	45	46	Mean $\pm$ Std	Item 95%	Tok.	maj@5	L	NC
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	94.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	95.0	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	94.4	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	94.0	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	91.2	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	91.0	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	89.8	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	91.2	2	21

4 Results

4.1 MATH-500 pass@1

Table 3 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. Llama AWQ-4 has the largest MATH-500 pass@1 drop among the Llama cells (−2.76 pp).

Table 4 is the test that matches RQ1. Llama AWQ-4 is the only MATH contrast whose interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). The other five MATH pass@1 contrasts are not significant, so a strict pass@1 order among the Qwen cells is not identified. An item bootstrap ($B = 10,000$) puts Qwen FP8 first on pass@1 in 80.3% of resamples and Llama FP8 first in 60.4%; Llama AWQ-4 is never first. No MATH contrast has its 90% CI inside $[-1, +1]$ pp. maj@5 accuracies are in Table 3: Qwen 94.4 / 95.0 / 94.4 / 94.0 and Llama 91.2 / 91.0 / 89.8 / 91.2 for BF16 / FP8 / AWQ-4 / GPTQ-4. Under majority voting the Qwen AWQ-4 point gap is gone (both 472/500, McNemar $p = 1$) and the Llama AWQ-4 point gap is about half as large (−2.76 to −1.4 pp). These are point-estimate changes: a paired item bootstrap of the change in the Llama gap (pass@1 minus maj@5) has an interval that includes zero (+1.36 [−0.24, +3.00] pp), so a statistical reduction is not shown. Majority voting also swaps Qwen AWQ-4 and GPTQ-4 relative to pass@1. McNemar p -values on maj@5 are all > 0.29 .

A two-way bootstrap that also resamples the generation seeds (Appendix Table 25) keeps the Llama AWQ-4 MATH-500 drop away from zero (−2.76 [−4.96, −0.68] pp), so that result does not rest on conditioning on the five seeds that were run. On GSM8K, with three seeds, the Llama AWQ-4 two-way interval is [−2.91, −0.15] pp and its upper end is close to zero, so the GSM8K drop is the weaker of the two. The paired standard error implied by the intervals of Table 4 is about 0.45–0.50 pp for Qwen and 0.60–0.70 pp for Llama, so a ± 1 pp TOST passes only when the point estimate is close to zero and is nearly unattainable for Llama. At 80% power the minimum detectable paired difference is 1.2–2.0 pp on MATH-500, 0.9–1.4 pp on GSM8K and 5.5–6.1 pp on GPQA-Diamond at $\alpha = 0.05$ (1.5–2.5, 1.2–1.7 and 6.8–7.6 pp at the first step of a Holm-6 family), so non-significant GPQA-Diamond contrasts say nothing about differences below about 7 pp. A graded alternative to the TOST, the posterior probability that a contrast lies within ± 1 pp, is in Appendix Table 31; for GPQA-Diamond it is at most 0.351.

4.2 GSM8K and GPQA-Diamond

Table 10 gives cell means; Tables 5 and 6 are the item-level tests. FP8 stays within 1.7 pp of BF16 on all three tasks in this grid (the largest point difference is Llama GPQA-Diamond, +1.68 pp, with a wide interval). Several GSM8K contrasts pass TOST at ± 1 pp, where $n = 1,319$ and the interval is narrower than on MATH-500. Llama AWQ-4 also drops on GSM8K (−1.57 pp, 95% CI [−2.55, −0.58], $p = 0.002$). Qwen AWQ-4 differs from Qwen BF16 by

Table 4: Primary: item-level bootstrap of MATH-500 pass@1 versus BF16 ($B = 10,000$). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts (primary family). TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST	McNemar p
Llama AWQ-4	-2.76	[-4.16, -1.44]	< 0.001	sig.	[-3.92, -1.64]	fail	0.296
Qwen AWQ-4	-0.88	[-1.88, +0.08]	0.069	n.s.	[-1.68, -0.08]	fail	1.000
Qwen GPTQ-4	-0.52	[-1.40, +0.36]	0.245	n.s.	[-1.28, +0.20]	fail	0.754
Qwen FP8	+0.40	[-0.48, +1.28]	0.383	n.s.	[-0.32, +1.12]	fail	0.581
Llama GPTQ-4	-0.32	[-1.56, +0.96]	0.619	n.s.	[-1.36, +0.72]	fail	1.000
Llama FP8	+0.28	[-0.92, +1.48]	0.647	n.s.	[-0.72, +1.28]	fail	1.000

Table 5: GSM8K item-level pass@1 versus matched BF16 ($n = 1,319$, 3 seeds). Holm over six contrasts. TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST
Llama AWQ-4	-1.57	[-2.55, -0.58]	0.002	sig.	[-2.38, -0.73]	fail
Qwen AWQ-4	-0.20	[-0.91, +0.48]	0.579	n.s.	[-0.78, +0.38]	pass
Qwen GPTQ-4	-0.13	[-0.86, +0.63]	0.729	n.s.	[-0.76, +0.51]	pass
Llama GPTQ-4	+0.28	[-0.53, +1.09]	0.511	n.s.	[-0.40, +0.96]	pass
Qwen FP8	+0.08	[-0.58, +0.73]	0.827	n.s.	[-0.45, +0.63]	pass
Llama FP8	+0.13	[-0.66, +0.94]	0.761	n.s.	[-0.53, +0.81]	pass

-5.56 pp on GPQA-Diamond (95% CI [-9.60, -1.52]). Its significance is borderline. With $B = 10,000$ the bootstrap p ranged from 0.0056 to 0.0104 across 20 resampling seeds, which puts the Holm-6 decision on both sides of α . With $B = 10^6$ the bootstrap p is 0.0081 (Holm-6 adjusted 0.049). The sign-flip permutation test gives $p = 0.011$ (adjusted 0.066) and the paired t -test gives $p = 0.009$ (adjusted 0.054). We therefore label it borderline rather than significant. It is not significant under the Holm-18 sensitivity (adjusted $p = 0.130$; Appendix Table 34). QRM draws the GPQA correct letter with `random.randint` after model load, which could have desynchronized the answer order across checkpoints. The stored campaign prompts show that it did not: within each family and seed, all four checkpoints saw byte-identical GPQA prompts (Appendix Section A). The contrast is therefore paired, but its significance is borderline and it is not a headline result.

4.3 Loops and near-cap completions

Across 88 runs the validator records **25** repetition-flagged completions (0.044% of 56,408). The observed maximum completion length is 32,737. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ). Three completions exceed 10,000 identical consecutive words. Near-cap completions ($\geq 32,500$ tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16’s 14.

Near-cap rows are very likely truncations. Of the 140 MATH-500 near-cap rows, 121 lie between 32,600 and 32,739 tokens, the band where 32,768 minus a typical prompt falls. Of the 57 GSM8K near-cap rows, 55 lie in that band. Only 24 MATH-500 rows and no GSM8K rows fall between the lowest possible cap (31,957) and 32,500. The stored campaign prompts give each row its own cap. Cap-band rows number 155 on MATH-500 (Qwen 14 / 16 / 27 / 29 and Llama 20 / 11 / 15 / 23 for BF16 / FP8 / AWQ-4 / GPTQ-4). Of the 140 near-cap rows, 139 are in the cap band, so the two heuristics agree closely; the cap band flags 16 further suspected truncations of long-prompt items. Neither heuristic is a native stop-reason label. Cap-band rows concentrate at difficulty levels 4 and 5 (129 of 155). On MATH-500, 120 of 134 unboxed rows are near-cap. A missing box on MATH is usually a long trace. It is not the 4-bit

Table 6: GPQA-Diamond pass@1 versus matched BF16 ($n = 198$, 3 seeds). Stored prompts confirm identical answer order across the four checkpoints of each family and seed (Appendix Section A). The Qwen AWQ-4 p uses $B = 10^6$ (other rows $B = 10,000$); it is borderline under Holm-6 (adjusted 0.049 bootstrap, 0.066 permutation) and not significant under Holm-18. It is not a headline claim. border.: borderline; n.s.: not significant.

Contrast	Δ (pp)	95% CI	p	Holm	90% CI	TOST
Qwen AWQ-4	-5.56	$[-9.60, -1.52]$	0.008	border.	$[-9.09, -2.19]$	fail
Qwen GPTQ-4	-2.36	$[-6.57, +1.85]$	0.288	n.s.	$[-6.06, +1.18]$	fail
Llama GPTQ-4	-1.18	$[-5.39, +3.03]$	0.573	n.s.	$[-4.71, +2.36]$	fail
Qwen FP8	-0.84	$[-4.71, +3.03]$	0.690	n.s.	$[-4.21, +2.53]$	fail
Llama AWQ-4	+0.84	$[-3.54, +5.05]$	0.711	n.s.	$[-2.86, +4.38]$	fail
Llama FP8	+1.68	$[-2.19, +5.39]$	0.403	n.s.	$[-1.52, +4.88]$	fail

accuracy gap: unboxed rows that are still scored correct are rare (at most two per seed file), and boxed rates are 98.96–99.68%. GSM8K is similar (52 of 66 unboxed rows are near-cap; boxed rates 99.70–99.87%). GPQA is different: 4,746 of 4,752 rows are unboxed, and only 12 are near-cap. GPQA extractive match is not a boxed-answer grade, and the long-tail proxy does not explain it. Native stop reasons were not logged, but vLLM stops generation at 32,768 minus the prompt length, and with the frozen tokenizers 147 of the 155 MATH-500 cap-band rows (and all 57 near-cap GSM8K rows) have exactly that completion length; 7 more lie within two tokens of it.

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 8, Figure 3), the tested Qwen AWQ-4/GPTQ-4 checkpoints showed +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Excluding near-cap pairs ($\geq 32,500$ tokens) the ratios remain +4.61%/+5.75%. Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

Figure 4 and Table 8 split paired token deltas by the 2×2 . Qwen AWQ-4 and GPTQ-4 Both-OK intervals exclude zero (+166 [+80, +254] and +136 [+45, +239]). Qwen FP8 and all three Llama Both-OK intervals include zero. The BF16-correct conditional Δ is positive with intervals excluding zero in every contrast (+166 to +417 tokens), including Qwen FP8, whose unconditional change is -4 tokens [-142, +134]. That column equals the n -weighted mix of Both-OK and BF16-only. For Qwen FP8, $(2297 \times 51 + 53 \times 5444)/2350 \approx 172$. Failure traces are long (15,843 vs. 3,256 tokens on Qwen BF16; 13,928 vs. 3,539 on Llama). The mirrored quant-correct conditional, the n -weighted mix of Both-OK and Quant-only, is -136 [-273, -3], +5 [-111, +114], +22 [-99, +151], -182 [-339, -35], -141 [-303, +12], and -183 [-335, -35] tokens. Three of those six intervals exclude zero, all on the negative side. Qwen AWQ-4 and GPTQ-4 stay near zero because the Both-OK lengthening offsets a smaller Quant-only tail.

The seed placebo (Table 7) sizes this selection effect directly. Applying the BF16-correct estimator to two BF16 seeds, with no quantization, gives +138 [+75, +207] tokens for Qwen and +235 [+151, +331] for Llama. The mirrored conditional is exactly the negative, by symmetry. Both FP8 BF16-correct values (+172 and +166) lie inside their family’s placebo interval, so this grid detects no FP8 excess over the placebo (paired excess +35 [-82, +158] for Qwen and -70 [-219, +74] for Llama). That is an absence of detected excess, not evidence that the two quantities are equal. The four 4-bit values (+326 to +417) exceed their placebo by about 100–250 tokens; the paired item-level intervals of that excess (shared resamples) exclude zero for Qwen AWQ-4 (+252 [+136, +380]), Qwen GPTQ-4 (+188 [+49, +335]) and Llama GPTQ-4 (+182 [+25, +344]) but not for Llama AWQ-4 (+97 [-66, +267]). That excess, not the raw

Table 7: Seed placebo for the correctness-conditioned length estimator on MATH-500. “Reference” is the arm whose correctness is conditioned on. Placebo rows compare BF16 seed s with BF16 seed $t \neq s$, pooled over 20 ordered pairs. Quantized rows repeat the BF16-correct column of Table 8. Item-level 95% CIs, $B = 10,000$.

Family	Contrast	Reference-correct Δ [95% CI]
Qwen-7B	BF16 vs BF16 (placebo)	+138 [+75, +207]
Qwen-7B	FP8 vs BF16	+172 [+63, +291]
Qwen-7B	AWQ-4 vs BF16	+390 [+261, +531]
Qwen-7B	GPTQ-4 vs BF16	+326 [+198, +464]
Llama-8B	BF16 vs BF16 (placebo)	+235 [+151, +331]
Llama-8B	FP8 vs BF16	+166 [+33, +302]
Llama-8B	AWQ-4 vs BF16	+332 [+182, +490]
Llama-8B	GPTQ-4 vs BF16	+417 [+261, +583]

conditional, is the part a quantization effect could explain. Under the independence assumption of Section 3.5, the excess estimates the unconditional change reweighted by per-item BF16 accuracy, which counts items that BF16 usually solves more heavily. A plug-in version that estimates the weights and the delta from the same BF16 seeds is biased upward; we therefore cross-fit, taking the weights from four BF16 seeds and the delta from the held-out seed, and averaging over the five folds. The cross-fitted reweighted change is +55 [−58, +171], +236 [+118, +361] and +197 [+72, +332] for Qwen FP8/AWQ-4/GPTQ-4 and −41 [−186, +100], +115 [−44, +281] and +160 [+10, +312] for Llama FP8/AWQ-4/GPTQ-4, close to the corresponding excess (+35, +252, +188; −70, +97, +182) and to the unconditional paired Δ for Qwen AWQ-4 (+236 vs. +254); it is smaller than the unconditional change for Qwen GPTQ-4 (+197 vs. +276), where the extra length falls on items that BF16 often fails, and the FP8 cells differ from their unconditional changes (−4 and −106) within wide intervals. Same-seed pairing does not couple the two arms: correctness agreement is 0.9536 for same-seed Qwen FP8/BF16 pairs and 0.9526 across seeds, with similar gaps in every contrast. The seed-matched 2×2 is therefore one arbitrary pairing, not a coupled design. We report both conditionals and the placebo, and we do not call any of them a reconciliation of prior length claims. Subtracting the placebo gives a diagnostic contrast that estimates an accuracy-weighted unconditional change, not a causal quantization effect. D (Table 9) remains a mismatch diagnostic.

The mean paired delta is also a poor summary of the typical trace. The median seed-averaged per-item delta is +50 [+10, +92] tokens for Qwen AWQ-4 and +4 [−43, +56] for Qwen GPTQ-4, against means of +254 and +276 (Appendix Table 26); the 10%-trimmed means are +118 and +92. All Llama medians have intervals that include zero. Most of the mean 4-bit length difference therefore comes from a minority of very long traces, so a mean-based “inflation” figure and a typical-trace figure answer different questions.

Mean tokens divided by pass@1 (Appendix Table 23) ranks Qwen FP8 first among Qwen cells. Campaign-length seconds, which divide by the measured tok/s, do not (Section 4.9).

4.5 Difficulty levels

A per-level breakdown needs each row’s MATH-500 `level1`. The compact JSON stores items in the campaign pipeline order, which differs from the public release order: a positional join matches the public gold in only 118 of 11,382 integer-gold rows (counted over the 20,000 MATH-500 completions: the 11,382 correct completions whose public-order gold answer is an integer, of which 118 have an extracted prediction equal to it). We therefore join on the exact question text in the stored campaign prompts; all 500 pipeline rows match distinct dataset items in both families. Table 11 reports seed-averaged pass@1 by level. The Llama AWQ-4 drop is

Table 8: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ : mean paired per-item token difference with an item-level 95% CI. 2×2 cells are mean quantized-BF16 Δ with an item-level 95% CI and pair count n . BF16-correct and quant-correct columns use the same item resamples on Both-OK $\cup$ BF16-only and Both-OK $\cup$ Quant-only.

Contrast	RoM	Mean Δ [95% CI]	Both OK	BF16 only	Quant only	Both wrong	BF16-correct	Quant-correct
Qwen FP8	-0.1%	-4 [-142, +134]	+51 ($n=2297$) [-42, +145]	+5444 ($n=53$) [+2922, +8035]	-6940 ($n=63$) [-9479, -4345]	+264 ($n=87$) [-1126, +1685]	+172 [+63, +291]	-136 [-273, -3]
Qwen AWQ-4	+6.3%	+254 [+116, +398]	+166 ($n=2272$) [+80, +254]	+6915 ($n=78$) [+4534, +9345]	-6533 ($n=56$) [-8933, -4255]	+894 ($n=94$) [-620, +2420]	+390 [+261, +531]	+5 [-111, +114]
Qwen GPTQ-4	+6.9%	+276 [+121, +431]	+136 ($n=2284$) [+45, +239]	+6892 ($n=66$) [+4090, +9896]	-4910 ($n=53$) [-8180, -1715]	+1893 ($n=97$) [+204, +3388]	+326 [+198, +464]	+22 [-99, +151]
Llama FP8	-2.3%	-106 [-284, +65]	+38 ($n=2121$) [-74, +152]	+2629 ($n=110$) [+1098, +4265]	-4159 ($n=117$) [-6040, -2428]	-968 ($n=152$) [-2252, +198]	+166 [+33, +302]	-182 [-339, -35]
Llama AWQ-4	+1.7%	+80 [-109, +268]	+91 ($n=2052$) [-25, +210]	+3090 ($n=179$) [+1995, +4286]	-4477 ($n=110$) [-6503, -2588]	-305 ($n=159$) [-1520, +907]	+332 [+182, +490]	-141 [-303, +12]
Llama GPTQ-4	+3.9%	+184 [+12, +359]	+66 ($n=2097$) [-43, +181]	+5915 ($n=134$) [+4335, +7605]	-4327 ($n=126$) [-6138, -2561]	+521 ($n=143$) [-506, +1585]	+417 [+261, +583]	-183 [-335, -35]

Table 9: Item-level mismatch excess $D = \bar{\Delta}_{\text{BF16-only}} - \bar{\Delta}_{\text{Both-OK}}$ on MATH-500. Both stratum means are formed on the same item resamples ($B = 10,000$; empty-stratum replicates skipped; none occurred). D is a diagnostic of correctness-conditioned mismatch asymmetry, not a causal quantization-induced token increase.

Contrast	Both-OK Δ	BF16-only Δ	Mismatch excess D [95% CI]
Qwen FP8	+51 [-42, +145]	+5444 [+2922, +8035]	+5394 [+2853, +7992]
Qwen AWQ-4	+166 [+80, +254]	+6915 [+4534, +9345]	+6748 [+4375, +9184]
Qwen GPTQ-4	+136 [+45, +239]	+6892 [+4090, +9896]	+6755 [+3938, +9786]
Llama FP8	+38 [-74, +152]	+2629 [+1098, +4265]	+2591 [+1047, +4244]
Llama AWQ-4	+91 [-25, +210]	+3090 [+1995, +4286]	+2999 [+1898, +4187]
Llama GPTQ-4	+66 [-43, +181]	+5915 [+4335, +7605]	+5849 [+4258, +7553]

Table 10: Cross-benchmark pass@1 (mean $\pm$ seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap summed over the three benchmarks (column totals 25 and 209).

Cell	MATH-500	GSM8K	GPQA-D	L / NC
Qwen BF16	94.00 $\pm$ 0.55	91.26 $\pm$ 0.29	50.34 $\pm$ 2.96	4 / 28
Qwen FP8	94.40 $\pm$ 1.05	91.33 $\pm$ 0.16	49.49 $\pm$ 1.52	0 / 24
Qwen AWQ-4	93.12 $\pm$ 0.67	91.05 $\pm$ 1.14	44.78 $\pm$ 3.04	5 / 32
Qwen GPTQ-4	93.48 $\pm$ 0.77	91.13 $\pm$ 0.27	47.98 $\pm$ 1.75	3 / 40
Llama BF16	89.24 $\pm$ 0.74	88.68 $\pm$ 0.46	46.13 $\pm$ 1.91	3 / 21
Llama FP8	89.52 $\pm$ 1.01	88.80 $\pm$ 0.62	47.81 $\pm$ 0.29	3 / 12
Llama AWQ-4	86.48 $\pm$ 1.96	87.11 $\pm$ 0.23	46.97 $\pm$ 2.02	4 / 21
Llama GPTQ-4	88.92 $\pm$ 1.55	88.96 $\pm$ 0.70	44.95 $\pm$ 4.32	3 / 31

not detectable at levels 1–2 and is larger at levels 3–5: -2.48 $[-4.95, -0.19]$, -3.75 $[-6.41, -1.09]$, and -4.63 $[-8.06, -1.34]$ pp at levels 3, 4, and 5. No other per-level interval excludes zero. The 30 per-level intervals are exploratory and not multiplicity-adjusted. A post hoc linear trend of the AWQ-minus-BF16 item difference on level is -1.25 $[-2.36, -0.15]$ pp per level for Llama AWQ-4 and -0.15 $[-0.96, +0.63]$ for Qwen AWQ-4; it assumes linearity in the level, is not multiplicity-adjusted, and the Llama interval nearly reaches zero. The serving subsets are unaffected, because they are identified by exact question text (Section 3.4).

4.6 Item-level descriptive checks

CPU-only reanalysis of the released compact JSON (`results/reports/item_level_descriptive_report.md`) is descriptive, not causal. MATH-500 has no item that is BF16-correct and quantized-wrong on all five seeds for any tested format. Incorrect MATH traces are longer than correct traces in every cell (for example, Qwen BF16 mean tokens 3,256 when correct vs. 15,843 when incorrect). GPQA-Diamond Qwen AWQ flips remain distributed (75 of 198 items flip on at least one seed; 0 items flip on all three seeds). These counts do not identify a small set of items that would explain the GPQA contrast.

4.7 Failure modes and the length signal

The compact JSON and the stored completions let us classify every incorrect MATH-500 completion as truncated, looping, unboxed, or a finished wrong answer (Appendix Table 29). The Llama AWQ-4 increase in errors is entirely finished wrong answers: $+3.04$ $[+1.68, +4.40]$ pp, against -0.24 $[-0.68, +0.16]$ pp for truncations. The Qwen 4-bit pass@1 gaps have the opposite composition. Truncations rise by $+0.52$ $[+0.08, +1.00]$ pp (AWQ-4; unadjusted $p = 0.013$,

Table 11: MATH-500 seed-averaged pass@1 (%) by difficulty level, joined to the public dataset by exact question text. n : items per level. Last rows: Llama AWQ-4 minus BF16 paired difference (pp) with an item-level 95% CI ($B = 10,000$). Per-level intervals are exploratory.

Cell	L1 $n=43$	L2 $n=90$	L3 $n=105$	L4 $n=128$	L5 $n=134$
Qwen BF16	98.1	98.9	97.9	93.6	86.7
Qwen FP8	98.1	98.2	98.1	94.5	87.6
Qwen AWQ-4	97.2	97.6	97.5	93.9	84.6
Qwen GPTQ-4	97.2	97.8	98.1	94.1	85.2
Llama BF16	95.3	94.7	95.0	90.5	77.9
Llama FP8	95.3	96.7	94.3	90.0	78.7
Llama AWQ-4	94.0	95.1	92.6	86.7	73.3
Llama GPTQ-4	95.8	95.8	95.4	88.6	77.3
Llama AWQ-4 – BF16	−1.40	+0.44	−2.48	−3.75	−4.63
95% CI	[−5.58, +1.40]	[−2.44, +3.33]	[−4.95, −0.19]	[−6.41, −1.09]	[−8.06, −1.34]

Holm-6 adjusted 0.08) and +0.48 [+0.04, +0.96] pp (GPTQ-4; unadjusted $p = 0.050$, adjusted 0.25), while the wrong-answer differences are +0.32 [−0.64, +1.28] and +0.04 [−0.88, +0.96] pp. These truncation intervals are among about two dozen class-by-contrast intervals, are unadjusted, and barely clear zero, so we read them as a small rise in truncations that accounts for most of the non-significant point gaps (−0.88 and −0.52 pp), not as an established effect. Apart from these, every class difference has an interval that includes zero or touches it at two decimals. In same-seed pairs, 179 BF16-correct completions are AWQ-4-incorrect for Llama against 110 in the other direction (net 69, the 2.76 pp drop), and 174 of those 179 are finished wrong answers; because same-seed pairing is an arbitrary coupling (Section 4.4), the marginal differences are the inferential quantity. The two 4-bit patterns therefore differ in kind: Llama AWQ-4 loses accuracy through reasoning errors in traces that finish, whereas the Qwen 4-bit point gaps are accounted for mostly by a few more traces that run into the output cap, which is also why their mean token difference is much larger than their median.

Shorter completions are more often correct (Appendix Table 30): the pooled AUROC is 0.83–0.94 across the eight cells. Most of that is item difficulty. Restricted to correct-versus-incorrect pairs within the same item, the AUROC is 0.62–0.76. No quantized checkpoint differs detectably from its BF16 baseline on the stratified AUROC (differences −0.05 to +0.04, every interval includes zero; Llama AWQ-4 −0.05 [−0.13, +0.03]), so these data show no detectable loss of the length signal under quantization, but only 48–134 items per cell have both outcomes and the intervals are wide. The one-sample length rule of Section 4.8 therefore works largely by separating easy items from hard ones.

Keyword counts of self-checking phrases are higher in the longer Qwen 4-bit and Llama GPTQ-4 traces even among finished completions, but not for Llama AWQ-4, whose accuracy loss is the robust one (Appendix Table 32); these are keyword counts, not an analysis of reasoning quality.

4.8 Observable modal agreement on MATH-500

Because MATH-500 was evaluated with five generations per item, we additionally examine whether observable answer agreement provides a useful gold-free abstention signal. Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration, and not as G-Pass@ k [17]. Table 14 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher observable answer agreement is associated with lower selective error. At 5/5 consensus, observed selective error is at most 0.27% across the evaluated configurations, but coverage ranges from

70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Several 0.00% risk cells have bootstrap intervals $[0, 0]$; Wilson 95% upper bounds on those $0/n$ cells are 0.82%–1.08%. Zero observed errors is not zero true error.

Llama AWQ-4 5/5 coverage is 70.2% versus 76.2% for BF16 (-6.0 pp, 95% CI $[-9.4, -2.6]$). Two further coverage intervals exclude zero and stop at -0.6 pp: Llama AWQ-4 at $\geq 4/5$ (-3.2 pp) and Qwen AWQ-4 at $\geq 4/5$ (-2.2 pp). The remaining coverage contrasts include zero. These 18 intervals are exploratory. They do not receive the Holm correction used for pass@1. Among 18 tests at $\alpha = 0.05$, about one false positive is expected. The Llama AWQ-4 5/5 interval is the one that sits well clear of zero. Appendix Table 19 lists the full pattern.

A one-sample length rule is also gold-free and costs one generation. For each seed we serve the shortest k traces, with k set to the published 5/5 coverage of that cell, and average the selective risk over the five seeds (Table 13). At that matched coverage the length rule’s risk is 1.61–4.56%, against at most 0.27% for 5/5 modal agreement. The five-sample rule still has the lower selective risk. It also spends about five times the tokens (T_5 below).

Table 12 gives the cost curve between those two ends. Unanimity of only two samples serves 82.1–92.7% of items at 0.69–2.36% risk. The length rule at the same coverage has 2.16–5.84% risk, so two samples cut risk by a factor of about 2.2–3.1 for twice the tokens. Three samples reach 0.18–1.07%. Every k beats the matched length rule in every cell. Most of the gain comes from the second sample, so a deployment that cannot afford five generations keeps most of the benefit at $k = 2$ or 3. These values come from the recovered answer strings, with equivalence recomputed by math-verify on the stored representations. The same code reproduces 22 of the 24 corresponding 5/5 coverage, 5/5 risk, and length-rule entries in Tables 14 and 13 exactly. The two exceptions share one cause: Qwen GPTQ-4 5/5 coverage is one item lower (86.6 vs. 86.8%) because equivalence is judged here on stored string representations rather than on the original parsed answers, and the matched-coverage length rule then keeps 433 instead of 434 items (1.62 vs. 1.61%). The reconstruction is computed by `scripts/analysis/revision_sensitivities.py`. Replacing math-verify equivalence by exact equality of the extracted answer strings lowers k -sample coverage by at most 0.4 pp and changes selective risk by at most 0.11 pp, so every comparison with the length rule is unchanged.

The fixed- k rules above always pay for k samples. An adaptive rule draws two samples, serves if they agree, and otherwise draws three more and serves the unique mode only if it has at least three of the five votes (Appendix Table 33; all 120 orderings of the five stored seeds, realised token cost). It reaches nearly the operating point of the fixed $\geq 3/5$ rule, with coverage 0.2 to 0.6 pp higher and selective risk 0.05 to 0.41 pp higher, at 37–46% fewer tokens (10.7–15.0k against 20.0–24.2k per item). A first-pair rule that serves as soon as any two of up to five samples agree serves 97–99% of items but at 2.2–6.4% risk, so it is not a useful operating point. These are point estimates on the items and seeds used throughout, with no fitted parameter; intervals are in the released JSON.

Mean T_5 is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. T_5 is five times the mean completion length. Abstained items still pay those tokens. Figure 1 shows the three operating points.

4.9 Measured serving throughput and Cost-of-Pass

This section does not claim a deployment cost ranking. Its finding is a mechanism: a few very long traces dominate the mean length of a small subset, so one draw of reasoning lengths can reorder cells that campaign-length costs order differently (Table 16). Table 15 is the supporting measurement; it separates tok/s from GPU-seconds per query. GPU-seconds equal output tokens divided by tok/s, so a long subset draw raises cost even when the kernel is faster. To size the effect we drew 20,000 random 20-prompt subsets stratified four per MATH level, gave each format one randomly chosen campaign seed, and held the measured Condition A tok/s fixed (Appendix Table 27). The campaign-length cheapest format is cheapest in only 58% of

Table 12: Cost curve for k -sample unanimity on MATH-500. Coverage (Cov.) and selective risk (Risk) are averaged over all $\binom{5}{k}$ seed subsets. Len.: one-sample shortest-trace rule at the same coverage, averaged over five seeds. Tokens per item are k times the mean completion length. The $k = 5$ column reproduces the 5/5 columns of Tables 14 and 13 except two entries that share one cause, a single item in Qwen GPTQ-4: coverage (86.6 here, 86.8 there) and the matched-coverage length-rule risk (1.62 here, 1.61 there). All values are percentages except tokens.

Model	Fmt	$k = 2$			$k = 3$			$k = 4$			$k = 5$		
		Cov.	Risk	Len.	Cov.	Risk	Len.	Cov.	Risk	Len.	Cov.	Risk	Len.
Qwen-7B	BF16	92.5	0.89	2.20	90.7	0.44	1.94	89.4	0.27	1.83	88.4	0.23	1.67
Qwen-7B	FP8	92.7	0.69	2.16	90.7	0.18	1.94	89.6	0.00	1.88	88.8	0.00	1.71
Qwen-7B	AWQ-4	91.2	1.05	2.54	88.8	0.43	2.34	87.4	0.27	2.15	86.4	0.23	1.99
Qwen-7B	GPTQ-4	91.6	0.87	2.31	89.3	0.29	1.92	87.8	0.09	1.69	86.6	0.00	1.62
Llama-8B	BF16	85.8	1.86	4.06	81.5	0.71	3.34	78.5	0.36	3.16	76.2	0.26	2.83
Llama-8B	FP8	86.9	1.91	4.47	82.9	0.63	3.82	80.3	0.20	3.39	78.2	0.00	3.17
Llama-8B	AWQ-4	82.1	2.36	5.84	76.8	1.07	4.90	73.1	0.44	4.71	70.2	0.00	4.56
Llama-8B	GPTQ-4	85.1	1.76	4.08	80.5	0.62	3.03	77.6	0.31	2.78	75.4	0.27	2.55
Tokens / item (Qwen; Llama)		$\approx 8.0-8.6\text{k}; 9.1-9.7\text{k}$			$\approx 12.0-12.9\text{k}; 13.7-14.5\text{k}$			$\approx 16.0-17.1\text{k}; 18.2-19.4\text{k}$			$\approx 20.0-21.4\text{k}; 22.8-24.2\text{k}$		

Table 13: One-sample length abstention at the published 5/5 coverage. For each seed, serve the shortest k traces. Risk is the mean over seeds 42–46. Modal risk is the 5/5 column of Table 14.

Model	Fmt	k	Length risk (%)	5/5 modal risk (%)
Qwen-7B	BF16	442	1.67	0.23
Qwen-7B	FP8	444	1.71	0.00
Qwen-7B	AWQ-4	432	1.99	0.23
Qwen-7B	GPTQ-4	434	1.61	0.00
Llama-8B	BF16	381	2.83	0.26
Llama-8B	FP8	391	3.17	0.00
Llama-8B	AWQ-4	351	4.56	0.00
Llama-8B	GPTQ-4	377	2.55	0.27

Table 14: Observable modal-answer agreement on MATH-500 ($n = 500$, seeds 42–46). Coverage is the fraction of items with a unique modal class of size $\geq k$. Risk is 1–selective accuracy among served items. Mean T_5 is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

Model	Fmt	$\geq 3/5$ cov.	$\geq 3/5$ risk	$\geq 4/5$ cov.	$\geq 4/5$ risk	5/5 cov.	5/5 risk	Mean T_5
Qwen-7B	BF16	96.0	1.67	93.4	0.43	88.4	0.23	20,057
Qwen-7B	FP8	96.6	1.66	92.6	0.00	88.8	0.00	20,038
Qwen-7B	AWQ-4	95.8	1.46	91.2	0.44	86.4	0.23	21,327
Qwen-7B	GPTQ-4	95.4	1.47	92.8	0.43	86.8	0.00	21,436
Llama-8B	BF16	94.0	2.98	87.8	0.68	76.2	0.26	23,283
Llama-8B	FP8	94.0	3.19	88.6	0.90	78.2	0.00	22,754
Llama-8B	AWQ-4	93.2	3.65	84.6	1.89	70.2	0.00	23,682
Llama-8B	GPTQ-4	93.8	2.77	86.4	0.46	75.4	0.27	24,203

draws for Qwen (GPTQ-4) and 62% for Llama (FP8), and the full four-format order equals the campaign-length order in 40% and 21% of draws. This simulation varies only the length draw and carries no timing or host uncertainty; it is the evidence for the mechanism.

The largest case among the measured subsets is Llama FP8 on Condition A. Within a repeat, output tokens equal tok/s times GPU-s/q; the three repeats each record $145,647/20 = 7,282.35$ tokens per query (the product of the printed means, 78.75×92.55 , gives 7,288 only because a mean of ratios is not a ratio of means). The campaign mean on the same 20 prompts is 4,889 tokens, and 4,550.8 on the full grid, which is 2.3% shorter than Llama BF16. Across the sixteen cell-by-condition comparisons of measured subset length with the campaign mean on the same prompts (Appendix Table 28) the ratio ranges from 0.88 to 1.49; Llama FP8 Condition A is the maximum (1.49) and Qwen FP8 Condition A is next (1.35). The latency percentiles of the 20 sequential requests (p90 of 19,334–20,314 and p95 of 26,489–27,135 tokens at the measured tok/s) imply about three requests of roughly 18,000 tokens or more, whereas the campaign reaches 15,000 tokens in 5.7% of Llama FP8 completions; under a beta-binomial count model, three or more such traces among 20 prompts has probability of at most about 4% (10% with a pooled rate). Resampling the five stored completions of those prompts exceeds 7,282 in only 19 of 20,000 draws, but five completions per prompt truncate the tail and overstate the rarity. We cannot attribute this excess. FP8 has the highest measured-to-campaign ratio in all four groups (1.35 and 1.07 for Qwen, 1.49 and 1.09 for Llama on Conditions A and B), the model files, quantization backend, prefix caching, chunked prefill and KV dtype are the same in both runs, and the campaign shows no elevated count of near-cap FP8 completions. What differs is single-stream versus batched scheduling, the seed, and the one extra input token per timer request (Section 3.4); the per-request lengths were not stored. A tail draw is possible but not established, and a single-BOS, multi-seed GPU rerun, which we have not run, would separate

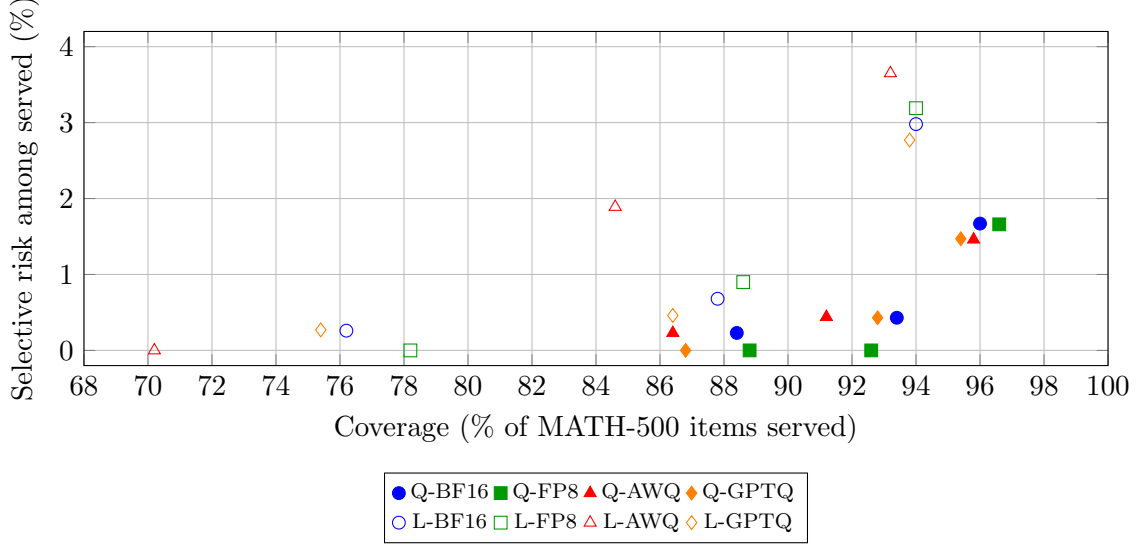


Figure 1: Three discrete MATH-500 operating points ($\geq 3/5$, $\geq 4/5$, $5/5$), not a continuous risk–coverage curve. Filled markers: Qwen-7B. Open markers: Llama-8B. Lower-right is the preferred corner (high coverage, low risk); the $\geq 3/5$ points sit at the upper right (high coverage, higher risk) and the $5/5$ points at the lower left (strict consensus, lower coverage).

the explanations. Condition A’s ranking of Llama BF16 as cheapest (\$0.0310 vs. about \$0.043) is therefore reported as a measurement of that subset, not as a stable ranking. Llama GPTQ-4 Condition B tok/s is 0.2% below BF16 (366.16 vs. 366.98) while hybrid cost is +24.9%; the gap is mostly length.

The brackets in Table 15 are the minimum and maximum hybrid cost across timing repeats, with campaign pass@1 held fixed. They are not a 95% interval. Repeats share seed 20260816, so they do not include length variance. Qwen BF16 Condition A tok/s runs only from 43.88 to 43.96. Qwen FP8 Condition B runs from 350.68 to 546.26, and its dollar min–max is \$0.0030–\$0.0047. Qwen FP8 Condition B tok/s of 449.79 in Table 15 is the mean of five ratios. Mean GPU-s/q is 8.64, and total tokens over total time is 432.5 tok/s, which is what the reported GPU-s/q and hybrid dollars imply. The campaign-length seconds below use total tokens over total time for every cell. Appendix Table 22 lists the five repeats. Llama FP8 and AWQ-4 Condition A means both round to \$0.0431. Their repeat ranges do not: \$0.0421–\$0.0449 and \$0.0423–\$0.0463.

Table 16 substitutes campaign mean tokens for the subset length and keeps the measured tok/s (total tokens over total time). Under that sensitivity, Qwen GPTQ-4 is cheapest on both conditions, so the AWQ/GPTQ reversal between A and B disappears. Llama FP8 is cheapest on both conditions and on mean tokens divided by pass@1. An item bootstrap of campaign tokens and pass@1, with tok/s held fixed ($B = 10,000$), gives Llama FP8 rank 1 on both campaign-length costs in all 10,000 draws. Qwen GPTQ-4 is rank 1 on Condition A in 99.9% of draws and on Condition B in 97.55%, with FP8 taking 2.45%; with the mean of per-repeat ratios, which overstates Qwen FP8’s tok/s, the Condition B split would be 59.6% against 40.4%. These resamples hold tok/s fixed, so they carry no timing uncertainty, and Qwen FP8’s Condition B tok/s varied by 56% across repeats (Appendix Table 22). Measured tok/s still depends on the subset’s length, so Table 16 is not a final serving measurement. Figure 2 plots the Condition B campaign-length seconds, not the subset dollars.

Qwen BF16 decodes at 43.92 tok/s on Condition A and 205.41 tok/s in the fixed-512 microbenchmark. Llama BF16, on the other host, reaches 72.34 and 362.01. The fixed-length gap remains, so it is not the Condition A length draw. The microbenchmark controls output length but still includes prefill, scheduling, and host overhead, so it does not isolate pure ker-

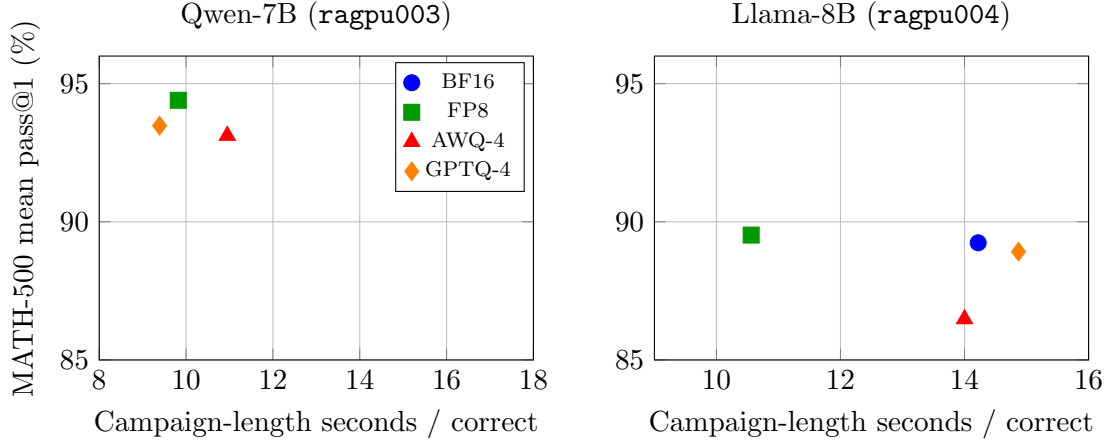


Figure 2: Campaign-length seconds per correct answer versus MATH-500 pass@1. The horizontal axis is full-grid mean tokens divided by the measured Condition B tok/s (total tokens over total time) and by pass@1. There are no timing whiskers: $R = 3$ or 5 does not support a percentile interval, and those repeats do not resample length. Points are not joined. Panels are separate hosts and are not a pooled trade-off plot.

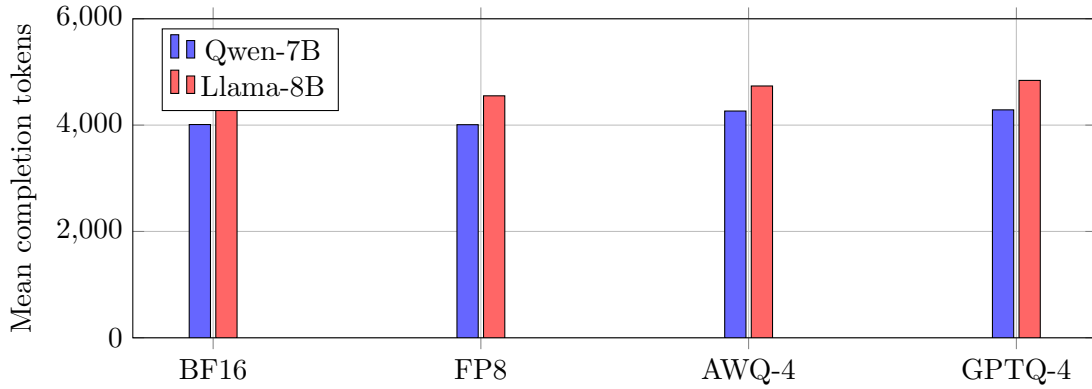


Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand). Bars are point estimates. Item-level 95% intervals on paired token Δ are in Table 8 and are not drawn on this axis.

nel decode speed. The Qwen BF16 baseline looks anomalous. Qwen-7B is the smaller model, yet it decodes about 40% slower than Llama-8B BF16. In the fixed-length microbenchmark FP8 speeds up Qwen by $2.13\times$ ($437.19/205.41$) but Llama by only $1.08\times$ ($392.05/362.01$), even though both use the same Marlin W8A16 kernel. Every Qwen Condition B saving in Table 15 (-36.0% to -45.9%) is relative to that baseline, which ran first on ragpu003. The Qwen FP8 Condition B repeats stepped up in run order (350.9, 350.7, 455.9, 545.2, 546.3 tok/s), which is consistent with changing runtime conditions, although host load is not identified as the cause. We therefore treat the Qwen savings as potentially confounded estimates whose bias direction is unknown; a controlled, interleaved same-node rerun would be needed to confirm them. Clocks were not logged. For AWQ-4 the backend is explained in Section 3.1. AWQ-Marlin is typically faster on A100 for compatible checkpoints, so every AWQ throughput number reflects the backend available to these artifacts as well as the format. That can contribute to Llama AWQ-4 decoding more slowly than BF16 on Condition A (70.20 vs. 72.34 tok/s). GPTQ-4 timing used `float16` while GPTQ-4 pass@1 used `bfloat16`, so the hybrid ratio joins two configurations. For AWQ-4 the timer also passes stored prompts that already include the `<think>` suffix, which the accuracy-campaign AWQ-4 prompts omit, so the AWQ-4 hybrid ratio likewise joins two con-

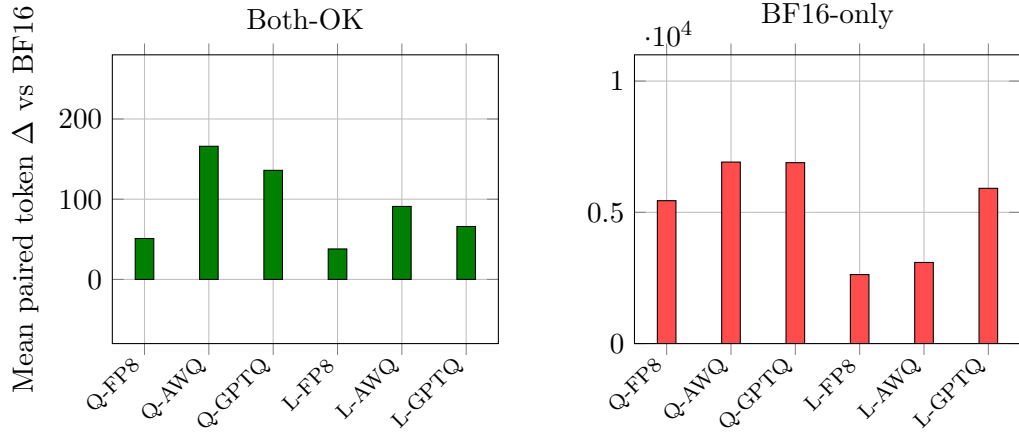


Figure 4: Paired token deltas by correctness, two panels so Both-OK is visible. Left: jointly-correct pairs (Qwen AWQ/GPTQ CIs exclude 0; FP8 and Llama CIs include 0; n in Table 8). Right: BF16-correct / quantized-wrong pairs. Item-level 95% CIs and n are in Table 8.

figurations. These FP8 numbers are A100 Marlin W8A16, not Hopper W8A8. Peak allocated VRAM is 53.9–56.1 GB, the 0.75 engine pool. Condition A latency is in Appendix Table 20.

Table 15: Hybrid cost on the controlled confirmation. Condition A: 20 sequential prompts. Condition B: 100-prompt `11m.generate`, `max_num_seqs=8`. Tok/s is mean $\pm$ sample SD. Brackets are the minimum and maximum hybrid cost across timing repeats, with campaign pass@1 held fixed. They are not a 95% interval. The B $\Delta\%$ column is the point estimate only. Qwen: `ragpu003`. Llama: `ragpu004`. GPTQ dollars join `float16` timing with `bfloat16` pass@1.

Model	Fmt	A tok/s	A GPU-s/q	A $\tilde{C}$ [min, max]	B tok/s	B GPU-s/q	B $\tilde{C}$ [min, max]	B $\Delta\%$
Qwen-7B	BF16	43.92 $\pm$ 0.04	102.51	\$0.0454 [0.0454, 0.0455]	252.72 $\pm$ 0.56	13.44	\$0.0060 [0.0059, 0.0060]	anchor
Qwen-7B	FP8	62.50 $\pm$ 0.08	84.60	\$0.0373 [0.0373, 0.0374]	449.79 $\pm$ 97.53	8.64	\$0.0038 [0.0030, 0.0047]	-36.0
Qwen-7B	AWQ-4	76.89 $\pm$ 0.70	48.04	\$0.0215 [0.0213, 0.0217]	418.15 $\pm$ 2.03	7.82	\$0.0035 [0.0035, 0.0035]	-41.3
Qwen-7B	GPTQ-4	82.67 $\pm$ 0.69	54.39	\$0.0242 [0.0241, 0.0245]	488.26 $\pm$ 0.64	7.23	\$0.0032 [0.0032, 0.0032]	-45.9
Llama-8B	BF16	72.34 $\pm$ 0.27	66.43	\$0.0310 [0.0309, 0.0311]	366.98 $\pm$ 1.90	10.13	\$0.0047 [0.0047, 0.0048]	anchor
Llama-8B	FP8	78.75 $\pm$ 2.89	92.55	\$0.0431 [0.0421, 0.0449]	481.42 $\pm$ 1.36	9.32	\$0.0043 [0.0043, 0.0044]	-8.2
Llama-8B	AWQ-4	70.20 $\pm$ 2.71	89.51	\$0.0431 [0.0423, 0.0463]	391.11 $\pm$ 0.48	11.63	\$0.0056 [0.0056, 0.0056]	+18.5
Llama-8B	GPTQ-4	63.49 $\pm$ 0.41	86.95	\$0.0407 [0.0405, 0.0410]	366.16 $\pm$ 0.79	12.60	\$0.0059 [0.0059, 0.0059]	+24.9

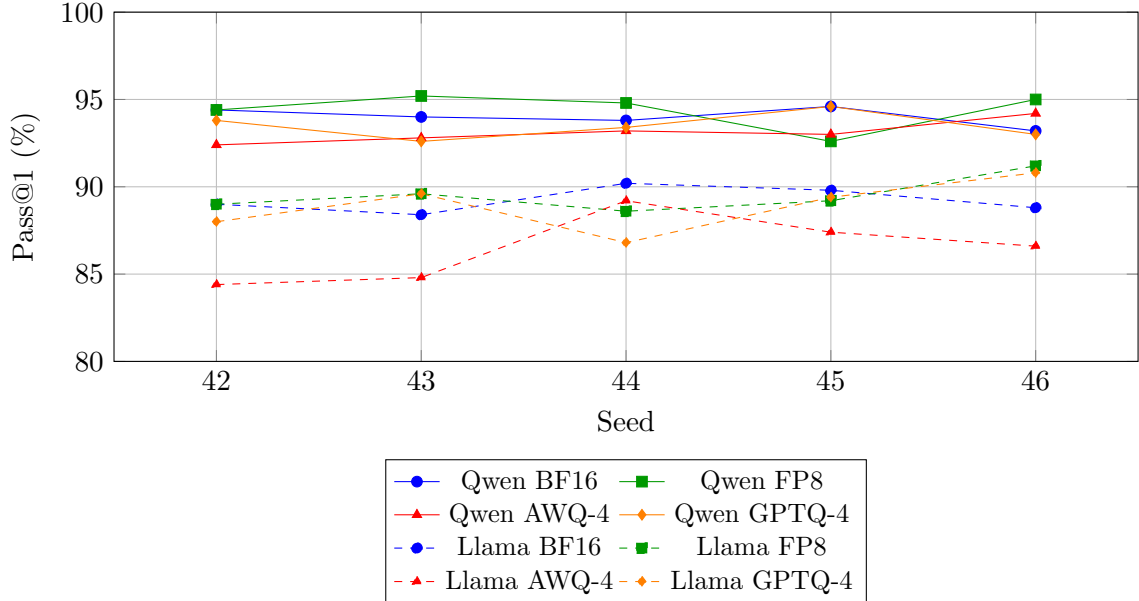


Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells.

Table 16: Campaign-length seconds per correct answer: full-grid mean MATH-500 tokens $\div$ measured tok/s $\div$ pass@1. Tok/s (the A and B columns) is total tokens over total time of the confirmation repeats and is held fixed; with the mean MATH-500 tokens and pass@1 of the campaign tables, the seconds can be recomputed from the printed values. Lower is cheaper. This is a sensitivity to subset length, not a replacement for Table 15.

Model	Fmt	A tok/s	A sec.	A rank	B tok/s	B sec.	B rank
Qwen-7B	BF16	43.92	97.16	4	252.72	16.89	4
Qwen-7B	FP8	62.50	67.93	3	432.54	9.82	2
Qwen-7B	AWQ-4	76.89	59.57	2	418.15	10.95	3
Qwen-7B	GPTQ-4	82.67	55.48	1	488.25	9.39	1
Llama-8B	BF16	72.33	72.14	2	366.98	14.22	3
Llama-8B	FP8	78.68	64.61	1	481.42	10.56	1
Llama-8B	AWQ-4	70.11	78.12	3	391.11	14.00	2
Llama-8B	GPTQ-4	63.49	85.74	4	366.16	14.87	4

5 Discussion

Each finding is a case in which the conclusion depends on the estimand: a single pass@1 versus an item-level interval and a class decomposition, a single conditional length versus a placebo-calibrated contrast, and a single serving subset versus the campaign length. Table 17 collects the per-cell values a practitioner would compare.

Llama AWQ-4 is the accuracy result that survives the item-level intervals on both MATH-500 and GSM8K. maj@5 lowers the Qwen AWQ-4 point gap to zero and the Llama AWQ-4 point gap by about half, but neither reduction is statistically established (the paired interval for the Llama change includes zero). The BF16-correct length column is positive in every contrast because it includes quantized failures, and a BF16-versus-BF16 placebo reproduces +138 to +235 tokens of it with no quantization (Table 7). The quant-correct column flips that sign in four of six contrasts, and three of those intervals exclude zero (Table 8). A correctness-conditioned length claim should therefore be reported alongside its seed placebo, as a diagnostic contrast that estimates an accuracy-weighted change rather than a corrected quantization effect.

Table 17: Practitioner summary. pass@1 and maj@5 are MATH-500 percent. Tokens are the campaign mean. A\$ and B\$ are $\tilde{C}_{\text{pass}}^{\text{hyb}}$. A sec. and B sec. are the campaign-length sensitivity. * marks the Holm-significant MATH or GSM8K pass@1 contrast.

Model	Fmt	pass@1	maj@5	Tokens	A \$	B \$	A / B sec.
Qwen-7B	BF16	94.00	94.4	4,011	0.0454	0.0060	97.2 / 16.9
Qwen-7B	FP8	94.40	95.0	4,008	0.0373	0.0038	67.9 / 9.8
Qwen-7B	AWQ-4	93.12	94.4	4,265	0.0215	0.0035	59.6 / 11.0
Qwen-7B	GPTQ-4	93.48	94.0	4,287	0.0242	0.0032	55.5 / 9.4
Llama-8B	BF16	89.24	91.2	4,657	0.0310	0.0047	72.1 / 14.2
Llama-8B	FP8	89.52	91.0	4,551	0.0431	0.0043	64.6 / 10.6
Llama-8B	AWQ-4*	86.48	89.8	4,736	0.0431	0.0056	78.1 / 14.0
Llama-8B	GPTQ-4	88.92	91.2	4,841	0.0407	0.0059	85.7 / 14.9

The placebo requires stochastic decoding; under greedy decoding it cannot be run, so whether the selection term carries over to greedy analyses is untested here.

Five-sample agreement has selective risk at most 0.27%, and a one-sample length rule at matched coverage is higher (Table 13). The gain does not require five samples. Two-sample unanimity already gives 2.2–3.1 times lower risk than the matched length rule at twice the tokens (Table 12).

Subset GPU-seconds and campaign-length seconds do not share a point order. Llama FP8 is cheapest on the campaign-length estimator in every item resample. The subset order depends on which prompts and which completions are drawn: random subsets reproduce the campaign-length order in only about a fifth to two fifths of draws, and Llama FP8’s Condition A realization is the most extreme of sixteen comparisons. Timing min–max does not cover that draw. AWQ throughput also depends on the backend available to these artifacts, and Qwen savings are relative to a BF16 baseline on `ragpu003` that looks anomalously slow. A single point order is not evidence that bit-width has a stable deployment ranking.

6 Limitations

1. **A100 W8A16 fallback.** FP8 checkpoints were executed as Marlin W8A16 on A100, not native W8A8. Numbers do not transfer to Hopper native W8A8.
2. **Measured serving scope.** Throughput and GPU-seconds are measured on balanced MATH-500 subsets, not the full campaign. Repeats share one sampling seed, so cost intervals omit length variance. Condition B elapsed time includes the straggler tail; occupancy time was not stored. $R = 3$ or 5 does not support a stable percentile interval. Eager execution was not compared with CUDA graphs. Qwen and Llama used different hosts. Formats ran sequentially on non-exclusive nodes, and clocks and co-tenant load were not logged. The Qwen BF16 baseline appears slow, which alone would bias the Qwen savings upward, but the quantized runs may also have been affected by changing conditions (the FP8 Condition B repeats step up in run order), so the direction of the net bias is unknown. The timer’s prompts carry one extra input token per request relative to the campaign, in every cell. The \$1.50/A100-hour rate is a pricing scenario. Peak VRAM is the 0.75 engine pool.
3. **Activation dtype.** AWQ-4 accuracy runs use `float16`. BF16, FP8, and accuracy-campaign GPTQ-4 use `bfloat16`. Serving GPTQ-4 uses `float16`. No BF16-at-`float16` control was run.
4. **Pathology metrology.** The loop detector is identical-word runs ≥ 20 . Phrase-level

cycles and `finish_reason` were not saved. Truncation is measured by a per-item cap band (within 16 tokens of the cap implied by the stored prompt), not by a native stop reason. GPQA letter order was a process-RNG draw; the stored prompts show it was identical across checkpoints and across the three generation seeds, so every GPQA number is conditional on one answer-letter permutation and position bias does not average out across seeds.

5. **Item order.** Compact JSON rows follow the campaign pipeline order. All paired analyses are within that order and are unaffected: the eight cells agree item by item. Joins against public per-item metadata (difficulty level) use exact question text from the stored campaign prompts; all 500 rows match distinct dataset items in both families.
6. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property. The one-sample length rule is a weak comparator; token-level confidence (mean log-probability, self-certainty) was not stored, so the stronger gold-free comparison is not made.
7. **Checkpoint provenance.** Both AWQ-4 checkpoints are `jaki`AJK uploads with unknown calibration. FP8 and GPTQ-4 are RedHatAI artifacts. The configurations suggest an AutoRound-style symmetric export in AWQ format (Section 3.1). Format, publisher, toolchain, activation dtype, AWQ kernel path, and chat template are collinear. No second AWQ artifact was quantized in this study.
8. **Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.
9. **Not evaluated.** This study does not evaluate AIME, LiveCodeBench, 32B models, Hopper native W8A8 FP8, energy measurements, production continuous batching, or probability calibration.
10. **Reproducibility surface.** Compact JSON, recovered MATH-500 answer strings, confirmation timing, and analysis reports are released. Full chain-of-thought traces are not. Compact JSON omits `finish_reason` and output token IDs. Replay needs an A100-80GB, the listed weights, and `runtime_manifest.json`. The serving records log driver 550.54.14; the accuracy-campaign job logs did not archive the driver. Each table and figure maps to its script and output file in `docs/PAPER_TO_CODE.md`. Analyses that need completion text (the reflection-marker counts, the frozen-tokenizer recount and exact-cap test, the `<think>` audit, and the cap-band assignment of the failure classes) run on the authors’ archive of completions, which is not released; every other analysis uses only the released records.
11. **Contamination.** MATH and GSM8K overlap public pretraining corpora. A paired within-item design reduces but does not eliminate contamination interaction with checkpoint.
12. **Launcher record.** Effective flags, including `max_model_len=32768`, are reconstructed in `runtime_manifest.json` from the launcher. The engine-initialization logs of the serving jobs confirm the engine version, eager mode, context length, seed, dtype and quantization setting per format; the accuracy-campaign logs contain no engine-initialization line, so its dtype rests on the launcher code, not on a log. The accuracy-campaign NVIDIA driver was not archived; the serving confirmation logs 550.54.14. Historical harness configs are not the campaign stack.

Threats to validity. Construct: pass@1, maj@5, token length, subset GPU-seconds, and campaign-length seconds are distinct estimands. The hybrid proxy is not Erol et al.’s per-problem Cost-of-Pass. Internal: Conditions A and B differ in subset and regime together; timing repeats do not resample length; Qwen and Llama used different hosts. External: two 7–8B families, one eager A100 stack, and three English benchmarks. Results do not transfer to CUDA-graph serving, Hopper W8A8, or a different publisher’s AWQ artifact.

7 Conclusion

Under this pinned A100 / vLLM 0.7.0 eager stack, the conclusions we can defend depend on the estimand: one robust accuracy drop, a placebo-sized selection effect in the BF16-correct length estimator, a gold-free agreement signal whose benefit arrives mostly with the second sample, and a subset serving cost that one length draw can reorder. Llama AWQ-4 falls -2.76 pp on MATH-500; maj@5 lowers the point estimate to about -1.4 pp (not a statistically established reduction), and the drop is concentrated at difficulty levels 3–5 and consists of finished wrong answers rather than truncations, whereas a small rise in truncations accounts for most of the non-significant Qwen 4-bit point gaps. Under sampled decoding, a BF16-versus-BF16 placebo reproduces $+138$ to $+235$ tokens of the BF16-correct length increase, and the quant-correct mirror does not keep its sign. Two-sample unanimity has 0.7–2.4% selective risk against 2.2–5.8% for a matched one-sample length rule; 5/5 agreement reaches at most 0.27%. Llama FP8 is cheapest once campaign length replaces the 20-prompt draw, and random 20-prompt subsets reproduce the campaign-length order in only 21–40% of draws. AWQ-4 results describe the tested `jakiAJK` artifacts, whose chat template, `float16` activations, and AWQ backend are confounded with the format. The GPQA AWQ-4 contrast is paired but remains borderline.

Reproducibility and Artifact Availability

The released artifact enables verification of reported analyses; reproducing the complete GPU campaign requires equivalent hardware and checkpoint availability.

CPU-reproducible from compact JSON: tables, analysis scripts, and `scripts/analysis/manuscript_sensitivities.py` (see `REPRODUCE.md`). Replay of the 88-run campaign requires an NVIDIA A100-80GB, the listed weights, and `results/reports/runtime_manifest.json`. The accuracy-campaign driver version was not archived (the serving records log 550.54.14). Full chain-of-thought traces are not released.

Code, patches, validation JSON, and analysis reports: [Manish06N/reasoning-compression-lab](#). Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`. Measured serving (confirmation): `results/reports/measured_serving_confirmation/`. First unconstrained timing (provenance only): `results/measured_serving/`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

Acknowledgment

PARAM Rudra HPC at IIT Patna (National Supercomputing Mission, C-DAC) provided compute.

CRedit authorship contribution statement

Manish Nandish: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Rajiv Misra:** Conceptualization, Resources, Supervision, Writing – review & editing. **Midhun-chakkaravarthy Janarthanan:** Supervision, Writing – review & editing.

The first author’s dual affiliation (IIT Patna and Lincoln University College) reflects a joint doctoral arrangement. Computational experiments, software, and formal analysis were performed by Manish Nandish. No experimental or software roles are claimed for Midhun-chakkaravarthy Janarthanan.

A Prompt

MATH-500 and GSM8K use the QRM template. Official `inference.py` calls `LightEval` with `use_chat_template=True`:

```
{question}
```

```
Please reason step by step, and put your final  
answer within \boxed{}
```

The prompt-cap recount applies that string through each family’s chat template, including the generation prompt. The six BF16, FP8, and GPTQ-4 templates end the generation prompt with `<think>\n`. The two AWQ-4 uploads end it at the assistant tag. For every item, benchmark, and seed, the AWQ-4 prompt equals the BF16 prompt minus that suffix, and FP8 and GPTQ-4 prompts are byte-identical to BF16. All 14,102 AWQ-4 completions open with exactly `<think>\n`, emitted by the model; no completion of the other six checkpoints does, because there the marker is part of the prompt. From the next token on, the AWQ-4 decoding context is identical to BF16’s, so the expected accuracy contrast is unaffected. AWQ-4 completion counts include the self-generated marker (2 tokens for both families in the frozen vLLM 0.7.0 runtime; a newer `tokenizers` release encodes the Llama marker as one token), which is below 0.05% of a mean MATH-500 trace. Evidence: `results/reports/q1_revision_analyses.json` and `results/reports/think_prefix_audit.json`.

GPQA-Diamond uses a different function in QRM’s `lighteval_custom/tasks/reasoning.py`. The prompt asks for a final line `Answer: $LETTER` and inserts the correct choice with `random.randint(0, 3)`. `inference.py` seeds Python’s RNG with the generation seed, but `LightEval` initializes the vLLM model before it builds the prompts, and code inspection shows that the vLLM worker then re-seeds Python’s RNG with the model-config seed (1234 by default). We did not reproduce the letters from that seed, but the stored prompts are consistent with an RNG state that does not depend on the generation seed: after normalizing the AWQ-4 template suffix, they are byte-identical across the four checkpoints of each family for every seed (594 of 594 per family), and also identical across the three generation seeds (198 of 198 items), so GPQA contrasts are paired and every GPQA number is conditional on one answer-letter permutation. The generation seed reaches vLLM only as a per-request sampling seed, so it changes sampling but not prompts. Stored GPQA rows agree with that template: 4,746 of 4,752 completions contain no `\boxed`, while `pass@1` is about 41–53% in every format, including BF16. Extractive match on GPQA is a letter match, not a boxed-answer grade. The QRM comparison row for GPQA in Table 1 compares answer-order draws from two different studies and is not a paired contrast.

Table 18: Public checkpoints and revisions used for the 56k campaign.

Cell	Hugging Face ID / revision
Qwen BF16	deepseek-ai/DeepSeek-R1-Distill-Qwen-7B 916b56a44061fd5cd7d6a8fb632557ed4f724f60
Qwen FP8	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic ceb2fcd178d477cd21b92ffb43164c74165a212c
Qwen AWQ-4	jakiAJK/DeepSeek-R1-Distill-Qwen-7B-AWQ 1d082b3ded4e7cefb6b6b32cd5e07c7d409832816
Qwen GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16 9559a91ed463eecd64b93a532fe5aa8cc25f2eb
Llama BF16	deepseek-ai/DeepSeek-R1-Distill-Llama-8B 6a6f4aa4197940add57724a7707d069478df56b1
Llama FP8	RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic 5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd
Llama AWQ-4	jakiAJK/DeepSeek-R1-Distill-Llama-8B-AWQ 57360e84a0e7278c6b3abb453f2dc812b213badb
Llama GPTQ-4	RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16 faa06ed0a4a5bdab76c4df12376e0c5228c0bdfe

B Checkpoints

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. Their `config.json` files carry `_name_or_path ./tmp_around_``d_awq`, which suggests an AutoRound export, and the model cards document no calibration, whereas the RedHatAI GPTQ-4 card documents `llm-compressor` with 2,048 calibration samples. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts. This study evaluates those public checkpoints under the dtypes in Table 2 and does not report Red Hat production serving measurements. Dataset revision SHAs are in `results/reports/runtime_manifest.json`.

Serving confirmation protocol and raw JSON live under `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md` and `results/measured_serving_confirmation/raw/`. Aggregates: `measured_serving_confirmation_report.json`. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`, `results/measured_serving/`) is provenance only and is not mixed with confirmation means.

C Reanalysis

Pathology, the item-level bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10,000$, seed 0.

D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix 23e9ead021111959; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses $B = 10,000$ problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence = $m_i/5$) are in the JSON report and are not treated as a primary calibration result.

T_5 is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90 T_5 are computed over all 500 items. Tokens per served query and tokens per

Table 19: Paired coverage Δ versus matched BF16 (percentage points). Intervals that exclude zero are marked *. Selective-accuracy Δ intervals are in the machine-readable modal report.

Contrast	$\geq 3/5$ cov. Δ [95% CI]	$\geq 4/5$ cov. Δ [95% CI]	$5/5$ cov. Δ [95% CI]
Qwen FP8	+0.6 [−0.8, +2.0]	−0.8 [−2.2, +0.6]	+0.4 [−1.6, +2.4]
Qwen AWQ-4	−0.2 [−1.8, +1.2]	−2.2 [−4.0, −0.6]*	−2.0 [−4.4, +0.2]
Qwen GPTQ-4	−0.6 [−2.0, +0.8]	−0.6 [−2.2, +1.0]	−1.6 [−4.2, +0.8]
Llama FP8	0.0 [−1.8, +1.8]	+0.8 [−1.6, +3.2]	+2.0 [−1.2, +5.2]
Llama AWQ-4	−0.8 [−2.8, +1.2]	−3.2 [−6.0, −0.6]*	−6.0 [−9.4, −2.6]*
Llama GPTQ-4	−0.2 [−2.2, +1.8]	−1.4 [−3.8, +1.0]	−0.8 [−4.2, +2.6]

Table 20: Confirmation single-stream serving (Condition A: 20 sequential prompts). Latency columns are means of per-run median/P90. $\tilde{C}_{\text{pass}}^{\text{hyb}}$ uses campaign MATH-500 pass@1 and a \$1.50/A100-hour scenario. Tok/s is mean $\pm$ sample SD.

Model	Fmt	tok/s	Med. lat (s)	P90 lat (s)	GPU-s/q	$\tilde{C}_{\text{pass}}^{\text{hyb}}$
Qwen-7B	BF16	43.92 $\pm$ 0.04	59.19	244.83	102.51	\$0.0454
Qwen-7B	FP8	62.50 $\pm$ 0.08	43.60	197.01	84.60	\$0.0373
Qwen-7B	AWQ-4	76.89 $\pm$ 0.70	26.47	110.43	48.04	\$0.0215
Qwen-7B	GPTQ-4	82.67 $\pm$ 0.69	34.78	67.51	54.39	\$0.0242
Llama-8B	BF16	72.34 $\pm$ 0.27	32.40	126.10	66.43	\$0.0310
Llama-8B	FP8	78.75 $\pm$ 2.89	38.00	253.55	92.55	\$0.0431
Llama-8B	AWQ-4	70.20 $\pm$ 2.71	40.12	246.14	89.51	\$0.0431
Llama-8B	GPTQ-4	63.49 $\pm$ 0.41	46.31	242.83	86.95	\$0.0407

correct served result use that same 500-item T_5 total in the numerator, so abstained queries are not free.

E Measured serving: latency, microbenchmark, FP8 repeats, and token proxy

Condition A latency detail (Table 20) is sequential decode of a balanced 20-prompt MATH-500 subset. Median latency is the mean of per-run medians. $R = 3$ except Llama AWQ-4 ($R = 5$; CV-triggered). Fixed-token microbenchmark (Table 21): 10 prompts $\times$ 512 tokens, $T = 0.0$, `ignore_eos`. Table 22 lists all five Qwen FP8 Condition B repeats. Table 23 reports mean tokens divided by pass@1. It is not measured wall-clock.

F Neighboring measurement studies

Table 24 is a compact comparison. It is not a claim that this paper replaces those studies.

G Additional sensitivities

These analyses are exploratory and post hoc. Table 25 checks whether the pass@1 contrasts depend on conditioning on the generation seeds that were run, and gives the minimum detectable difference. Table 26 shows how much of the mean token difference survives a median and a trimmed mean. Table 27 sizes the Condition A length-draw effect by simulation and Table 28 compares every measured subset length with the campaign mean on the same prompts. Tables 29 and 30 give the failure-class and length-AUROC analyses of Section 4.7; Tables 31, 32 and 33 give graded equivalence, reflection-marker counts, and adaptive agreement sampling.

Table 21: Fixed-token decode microbenchmark (10 prompts $\times$ 512 tokens). Fixes the output length, so it removes trace-length variation; it still includes prefill, scheduling, and host overhead, so it does not isolate kernel decode speed. Secondary to task-realistic Conditions A/B.

Model	Fmt	Raw decode tok/s
Qwen-7B	BF16	205.41
Qwen-7B	FP8	437.19
Qwen-7B	AWQ-4	377.73
Qwen-7B	GPTQ-4	401.19
Llama-8B	BF16	362.01
Llama-8B	FP8	392.05
Llama-8B	AWQ-4	344.04
Llama-8B	GPTQ-4	302.54

Table 22: Qwen-7B FP8 Condition B: five technical repeats with identical aggregate 373,794 output tokens. The tok/s column is a ratio. The mean GPU-s/q of 8.64 implies $3,738/8.64 \approx 432.5$ tok/s, which is not the mean of the five ratios (449.79). Slow: tok/s < 400. Fast: tok/s > 500. The cause of the timing regimes was not identified.

Rep	tok/s	GPU-s/q	$\tilde{C}_{\text{pass}}^{\text{hyb}}$	Regime
1	350.91	10.6522	\$0.0047	slow
2	350.68	10.6591	\$0.0047	slow
3	455.90	8.1990	\$0.0036	mid
4	545.18	6.8563	\$0.0030	fast
5	546.26	6.8427	\$0.0030	fast
Mean of ratios	449.79 ± 97.53	8.64	\$0.0038	keep all five
Median / [Q1, Q3]	455.90 / [350.91, 545.18]	—	—	—

Most use only the released compact records; those that need completion text run on the authors’ archive of completions (Limitations, item on the reproducibility surface).

Graded equivalence. The TOST at ± 1 pp fails for every MATH-500 contrast, which does not say how close the evidence comes. Under a flat Dirichlet Bayesian bootstrap over the paired item deltas (Table 31), the posterior probability that the contrast lies within ± 1 pp is 0.91 and 0.87 for Qwen and Llama FP8, 0.86 and 0.84 for GPTQ-4, and only 0.61 for Qwen AWQ-4; for Llama AWQ-4 it is 0.006. On GSM8K it is 0.96 to 1.00 for every contrast except Llama AWQ-4 (0.13). On GPQA-Diamond it is at most 0.351, so those contrasts are uninformative about equivalence. These probabilities grade the evidence under the archived seeds; they are not a hypothesis test and do not license an equivalence claim.

Reflection markers. A keyword count of self-checking phrases (“wait”, “hmm”, “alternatively”, check and verify phrases; Appendix Table 32) is higher in the longer 4-bit traces even among finished completions only (a comparison that conditions on finishing, so it does not describe the truncated traces): +0.71 [+0.53, +0.89] markers per 1,000 tokens for Qwen AWQ-4, +0.91 [+0.68, +1.14] for Qwen GPTQ-4 and +0.69 [+0.43, +0.94] for Llama GPTQ-4, whereas Llama AWQ-4, whose accuracy loss is the robust one, does not differ (+0.05 [−0.20, +0.30]). The pooled Qwen FP8 decrease (−0.46) disappears among finished completions (−0.07 [−0.27, +0.14]), so it reflected truncated traces. These are keyword counts, not an analysis of reasoning quality.

Table 23: MATH-500 mean tokens divided by pass@1. This is the order a shared throughput would produce. It is not measured wall-clock. Primary cost evidence is Table 15 and Table 16.

Cell	Mean tok.	Tokens / correct
Qwen BF16	4,011.4	4,267
Qwen FP8	4,007.7	4,245
Qwen AWQ-4	4,265.3	4,580
Qwen GPTQ-4	4,287.2	4,586
Llama BF16	4,656.6	5,218
Llama FP8	4,550.8	5,084
Llama AWQ-4	4,736.5	5,477
Llama GPTQ-4	4,840.5	5,444

Table 24: Neighboring measurement studies. In the engine column, a named version means that serving stack is frozen while checkpoints vary. Length estimands and serving columns are what each paper reports as primary evidence, not an exhaustive inventory.

Study	Models; bits	Engine; seeds	Length estimand; correctness split	Serving / cost; statistics
This paper	2 R1 distill (7B/8B); BF16, FP8, AWQ-4, GPTQ-4	vLLM 0.7.0 eager, A100; 5 seeds MATH, 3 GSM/GPQA	RoM, paired Δ , both conditionals, seed placebo; 2×2 + both mirrors	GPU-s, campaign-length seconds; item-level bootstrap, Holm, TOST
QRM [18]	R1 distill family; W/A/KV incl. W4A16	QRM harness (multi-config); 3 seeds (v2)	mean tokens; not 2×2	limited; per-cell means
Lian et al. [14]	reasoning models; low-bit PTQ	paper stack; greedy ($T = 0$), 100 questions per benchmark, one seed	CTIR (ratio of mean CoT-only lengths); BF16-correct vs rest (suppl.)	token cost; length ratio
Lotfi et al. [19]	1.5B–32B reasoning; aggressive PTQ	paper stack; paper-specific seeds	CoT length, overthinking; abandoned intermediates	logit-penalty Pareto; error-type counts
Kurtic et al. [11]	Llama-3.1 + R1-distill; FP8/INT8/INT4	vLLM, several GPUs; large evals	not CoT-primary; N/A	sync vs continuous batching; accuracy + tok/s
Helcig et al. [8]	Llama/Qwen (SLQ); 3.3–6 bpp	vLLM kernels; 3 seeds (GSM)	thinking tokens; TL vs DL fidelity	vLLM throughput; EAR, Wilcoxon
Erol et al. [3]	general LMs; N/A	N/A	N/A	C_{pass} definition; economic
Hochlehnert et al. [10]	reasoning LMs; N/A	deliberately unpinned; many seeds	N/A	N/A; reproducibility

Table 25: Two-way (items $\times$ seeds) bootstrap of the pass@1 contrasts of Tables 4, 5 and 6 ($B = 10,000$), and the minimum detectable paired difference (MDE) at 80% power: un-adjusted (two-sided $\alpha = 0.05$, multiplier $2.80\times$ the paired item-level SE) and at the first step of a Holm-6 family ($\alpha = 0.05/6$, multiplier $3.48\times$). Exploratory and post hoc; from `scripts/analysis/review_additions.py`.

Bench.	Contrast	Δ (pp)	Two-way 95% CI	MDE (pp)	MDE, Holm-6 (pp)
MATH-500	Qwen FP8	+0.40	$[-1.28, +2.00]$	1.25	1.55
MATH-500	Qwen AWQ-4	-0.88	$[-2.44, +0.72]$	1.39	1.73
MATH-500	Qwen GPTQ-4	-0.52	$[-1.80, +0.72]$	1.27	1.58
MATH-500	Llama FP8	+0.28	$[-1.68, +2.32]$	1.69	2.10
MATH-500	Llama AWQ-4	-2.76	$[-4.96, -0.68]$	1.97	2.45
MATH-500	Llama GPTQ-4	-0.32	$[-2.76, +2.00]$	1.79	2.22
GSM8K	Qwen FP8	+0.08	$[-0.76, +0.96]$	0.93	1.16
GSM8K	Qwen AWQ-4	-0.20	$[-1.82, +1.31]$	1.00	1.24
GSM8K	Qwen GPTQ-4	-0.13	$[-1.09, +0.81]$	1.05	1.30
GSM8K	Llama FP8	+0.13	$[-1.31, +1.52]$	1.14	1.42
GSM8K	Llama AWQ-4	-1.57	$[-2.91, -0.15]$	1.40	1.74
GSM8K	Llama GPTQ-4	+0.28	$[-0.86, +1.44]$	1.18	1.47
GPQA-D	Qwen FP8	-0.84	$[-7.41, +5.56]$	5.65	7.02
GPQA-D	Qwen AWQ-4	-5.56	$[-11.11, +0.00]$	5.90	7.33
GPQA-D	Qwen GPTQ-4	-2.36	$[-9.43, +4.88]$	6.15	7.64
GPQA-D	Llama FP8	+1.68	$[-4.04, +7.07]$	5.50	6.83
GPQA-D	Llama AWQ-4	+0.84	$[-5.56, +7.07]$	6.11	7.59
GPQA-D	Llama GPTQ-4	-1.18	$[-8.08, +5.56]$	6.03	7.49

Table 26: MATH-500 paired token difference (quantized - BF16) of the seed-averaged per-item delta: mean, median, and 10%-trimmed mean, each with an item-level 95% CI ($B = 10,000$). The means reproduce Table 8 up to Monte Carlo error. Exploratory; from `scripts/analysis/review_additions.py`.

Contrast	Mean	Median	10%-trimmed mean
Qwen FP8	-4 $[-144, +137]$	0 $[-33, +48]$	+30 $[-34, +92]$
Qwen AWQ-4	+254 $[+116, +394]$	+50 $[+10, +92]$	+118 $[+49, +192]$
Qwen GPTQ-4	+276 $[+123, +433]$	+4 $[-43, +56]$	+92 $[+27, +169]$
Llama FP8	-106 $[-278, +67]$	-29 $[-105, +9]$	-57 $[-148, +32]$
Llama AWQ-4	+80 $[-110, +270]$	+11 $[-61, +83]$	+79 $[-26, +183]$
Llama GPTQ-4	+184 $[+11, +359]$	+38 $[-19, +116]$	+90 $[-20, +205]$

Table 27: Subset-draw simulation for Condition A. Each of 20,000 draws takes a random 20-prompt MATH-500 subset stratified four per difficulty level, assigns each format one randomly chosen campaign seed, and costs it as subset mean tokens $\div$ measured Condition A tok/s $\div$ campaign pass@1. Columns: probability that each format is cheapest (bold: the campaign-length cheapest format of Table 16) and probability that the full four-format order equals the campaign-length order. Only the length draw varies; tok/s is held fixed, so no timing or host uncertainty is included. From `scripts/analysis/review_additions.py`.

Family	P(cheapest)				P(order = campaign)
	BF16	FP8	AWQ-4	GPTQ-4	
Qwen-7B	0.000	0.078	0.343	0.580	0.403
Llama-8B	0.243	0.616	0.104	0.037	0.206

Table 28: Measured subset length against the campaign. Each entry is the mean output tokens per query measured in the serving confirmation / the campaign mean (five seeds) over the same prompts (ratio in parentheses). Condition A: 20 prompts; Condition B: 100 prompts. Llama FP8 Condition A has the largest ratio of the sixteen comparisons. From `scripts/analysis/followup_checks.py`.

Cell	Condition A	Condition B
Qwen BF16	4,502 / 3,864 (1.17)	3,396 / 3,613 (0.94)
Qwen FP8	5,288 / 3,903 (1.35)	3,738 / 3,481 (1.07)
Qwen AWQ-4	3,694 / 3,791 (0.97)	3,270 / 3,703 (0.88)
Qwen GPTQ-4	4,496 / 4,706 (0.96)	3,530 / 3,662 (0.96)
Llama BF16	4,806 / 5,289 (0.91)	3,716 / 4,047 (0.92)
Llama FP8	7,282 / 4,889 (1.49)	4,489 / 4,131 (1.09)
Llama AWQ-4	6,275 / 5,473 (1.15)	4,550 / 4,391 (1.04)
Llama GPTQ-4	5,521 / 5,420 (1.02)	4,614 / 4,357 (1.06)

Table 29: Change in the incorrect-completion classes relative to BF16 on MATH-500 (percentage points of all completions, seed-averaged per item, item-level 95% CI, $B = 10,000$). Every incorrect completion has exactly one class, in priority order: *truncated* (within 16 tokens of its per-item output cap), *loop* (identical-word run ≥ 20), *no box* (finished, no `\boxed`), *wrong* (finished, boxed, wrong answer). The four classes sum to the all-errors column, which equals minus the pass@1 difference. The loop and no-box differences are omitted because they are near zero: at most 0.08pp in magnitude, with all intervals inside $[-0.12, +0.20]$ (full values in `results/reports/trace_failure_analysis.json`). Truncation is the cap-band heuristic of Section 4.3, not a native stop reason. Exploratory; from `scripts/analysis/trace_failure_analysis.py`.

Contrast	Truncated	Wrong answer	All errors
Qwen FP8	+0.08 $[-0.24, +0.40]$	-0.52 $[-1.32, +0.28]$	-0.40 $[-1.28, +0.48]$
Qwen AWQ-4	+0.52 $[+0.08, +1.00]$	+0.32 $[-0.64, +1.28]$	+0.88 $[-0.04, +1.88]$
Qwen GPTQ-4	+0.48 $[+0.04, +0.96]$	+0.04 $[-0.88, +0.96]$	+0.52 $[-0.36, +1.40]$
Llama FP8	-0.36 $[-0.80, +0.04]$	+0.12 $[-1.04, +1.28]$	-0.28 $[-1.48, +0.88]$
Llama AWQ-4	-0.24 $[-0.68, +0.16]$	+3.04 $[+1.68, +4.40]$	+2.76 $[+1.40, +4.12]$
Llama GPTQ-4	+0.12 $[-0.44, +0.64]$	+0.16 $[-1.08, +1.44]$	+0.32 $[-0.92, +1.56]$

Table 30: Does a shorter completion predict a correct one (MATH-500, 5 seeds)? Pooled AUROC uses all 2,500 completions of a cell. Item-stratified AUROC uses only correct-versus-incorrect pairs within the same item, which removes item difficulty; its quantized-minus-BF16 difference uses shared item resamples. Item-level 95% CIs. The last column counts items with at least one correct and one incorrect completion. Exploratory; from `scripts/analysis/trace_failure_analysis.py`.

Cell	Pooled AUROC	Stratified AUROC	Stratified – BF16	Items
Qwen BF16	0.936 $[0.91, 0.96]$	0.737 $[0.63, 0.84]$	—	48
Qwen FP8	0.927 $[0.89, 0.96]$	0.714 $[0.62, 0.81]$	-0.023 $[-0.14, +0.09]$	48
Qwen AWQ-4	0.925 $[0.90, 0.95]$	0.697 $[0.60, 0.78]$	-0.040 $[-0.17, +0.08]$	59
Qwen GPTQ-4	0.930 $[0.90, 0.96]$	0.763 $[0.68, 0.84]$	+0.027 $[-0.10, +0.16]$	56
Llama BF16	0.875 $[0.84, 0.91]$	0.675 $[0.61, 0.74]$	—	107
Llama FP8	0.871 $[0.82, 0.91]$	0.664 $[0.59, 0.73]$	-0.011 $[-0.10, +0.07]$	91
Llama AWQ-4	0.830 $[0.79, 0.87]$	0.624 $[0.56, 0.68]$	-0.051 $[-0.13, +0.03]$	134
Llama GPTQ-4	0.901 $[0.87, 0.93]$	0.717 $[0.65, 0.78]$	+0.042 $[-0.04, +0.12]$	110

Table 31: Graded equivalence of the pass@1 contrasts: posterior probability that the contrast lies within ± 1 pp or ± 2 pp of zero, and that the drop exceeds 1 pp, under a flat Dirichlet Bayesian bootstrap over the paired item deltas ($B = 10,000$). Conditional on the archived seeds. These grade the evidence; they are not a hypothesis test and do not replace the TOST of Tables 4–6. Exploratory; from `scripts/analysis/adaptive_reflection_rope.py`.

Bench.	Contrast	Δ (pp)	$P(\Delta < 1)$	$P(\Delta < 2)$	$P(\Delta < -1)$
MATH-500	Qwen FP8	+0.40	0.912	1.000	0.002
MATH-500	Qwen AWQ-4	−0.88	0.608	0.985	0.392
MATH-500	Qwen GPTQ-4	−0.52	0.861	0.999	0.138
MATH-500	Llama FP8	+0.28	0.867	0.997	0.018
MATH-500	Llama AWQ-4	−2.76	0.006	0.133	0.994
MATH-500	Llama GPTQ-4	−0.32	0.844	0.995	0.139
GSM8K	Qwen FP8	+0.08	0.996	1.000	0.001
GSM8K	Qwen AWQ-4	−0.20	0.987	1.000	0.012
GSM8K	Qwen GPTQ-4	−0.13	0.988	1.000	0.010
GSM8K	Llama FP8	+0.13	0.979	1.000	0.004
GSM8K	Llama AWQ-4	−1.57	0.126	0.806	0.875
GSM8K	Llama GPTQ-4	+0.28	0.958	1.000	0.001
GPQA-D	Qwen FP8	−0.84	0.351	0.642	0.469
GPQA-D	Qwen AWQ-4	−5.56	0.012	0.042	0.987
GPQA-D	Qwen GPTQ-4	−2.36	0.205	0.416	0.737
GPQA-D	Llama FP8	+1.68	0.278	0.534	0.084
GPQA-D	Llama AWQ-4	+0.84	0.324	0.604	0.203
GPQA-D	Llama GPTQ-4	−1.18	0.315	0.582	0.531

Table 32: Reflection-marker counts in the stored MATH-500 completions (regular expressions for “wait”, “hmm”, “alternatively”, and check/verify phrases; keyword counts, not a semantic analysis). Columns: all markers per 1,000 tokens; the same for correct / incorrect finished (non-truncated) completions; and the quantized-minus-BF16 difference per 1,000 tokens, pooled over all completions and over finished (non-truncated) completions only, with item-level 95% CIs ($B = 10,000$). Exploratory; from `scripts/analysis/adaptive_reflection_rope.py`.

Cell	All /1k	Correct / incorrect /1k	− BF16, pooled	− BF16, finished only
Qwen BF16	7.07	6.12 / 9.21	—	—
Qwen FP8	6.61	6.04 / 9.57	−0.46 [−0.99, −0.04]	−0.07 [−0.27, +0.14]
Qwen AWQ-4	7.76	6.91 / 9.51	+0.69 [+0.09, +1.33]	+0.71 [+0.53, +0.89]
Qwen GPTQ-4	8.60	7.80 / 10.80	+1.52 [+0.53, +2.75]	+0.91 [+0.68, +1.14]
Llama BF16	9.43	8.84 / 10.92	—	—
Llama FP8	9.44	8.84 / 11.30	+0.01 [−0.29, +0.30]	+0.09 [−0.16, +0.33]
Llama AWQ-4	9.36	8.96 / 10.55	−0.06 [−0.38, +0.24]	+0.05 [−0.20, +0.30]
Llama GPTQ-4	10.30	9.28 / 12.07	+0.88 [+0.43, +1.35]	+0.69 [+0.43, +0.94]

Table 33: Adaptive agreement sampling on MATH-500. Each entry is coverage (%) / selective risk (%) / mean tokens per item (thousands). Fixed $\geq 3/5$: the five-sample rule of Table 14. Adaptive: draw two samples and serve if they agree; otherwise draw three more and serve the unique mode only if it has at least three of the five votes. Fixed $k=2$: unanimity of two samples (Table 12). Adaptive and fixed $k=2$ are averaged over all 120 orderings of the five stored seeds, with the realised tokens of the samples drawn; item-level bootstrap intervals are in `results/reports/adaptive_reflection_rope.json`. No parameter was fitted to these data. From `scripts/analysis/adaptive_reflection_rope.py`.

Cell	Fixed $\geq 3/5$	Adaptive	Fixed $k=2$
Qwen BF16	96.0 / 1.67 / 20.1	96.3 / 1.72 / 10.9	92.5 / 0.89 / 8.0
Qwen FP8	96.6 / 1.66 / 20.0	96.8 / 1.71 / 10.7	92.7 / 0.69 / 8.0
Qwen AWQ-4	95.8 / 1.46 / 21.3	96.2 / 1.77 / 11.9	91.2 / 1.05 / 8.5
Qwen GPTQ-4	95.4 / 1.47 / 21.4	95.7 / 1.65 / 11.9	91.6 / 0.87 / 8.6
Llama BF16	94.0 / 2.98 / 23.3	94.5 / 3.34 / 14.0	85.8 / 1.86 / 9.3
Llama FP8	94.0 / 3.19 / 22.8	94.5 / 3.47 / 13.3	86.9 / 1.91 / 9.1
Llama AWQ-4	93.2 / 3.65 / 23.7	93.6 / 4.06 / 15.0	82.1 / 2.36 / 9.5
Llama GPTQ-4	93.8 / 2.77 / 24.2	94.4 / 3.18 / 14.7	85.1 / 1.76 / 9.7

H Holm-18 sensitivity

Primary multiplicity control is Holm–Bonferroni within each benchmark’s six format contrasts. Table 34 applies Holm jointly to all 18 contrasts as a stricter sensitivity. Llama MATH AWQ-4 and Llama GSM8K AWQ-4 remain significant. Qwen GPQA AWQ-4 does not (Holm-18 $p = 0.130$, from the $B = 10^6$ bootstrap $p = 0.0081$); it is already borderline under Holm-6.

Table 34: Secondary Holm correction over all 18 pass@1 contrasts. Primary inference remains Holm-6 within each benchmark. sig.: Holm-significant at $\alpha = 0.05$; border.: borderline (see Section 3.5); n.s.: not significant.

Bench.	Contrast	p	Holm-6	Holm-18 p	Holm-18
MATH	Llama AWQ-4	< 0.001	sig.	< 0.001	sig.
MATH	Qwen AWQ-4	0.069	n.s.	1.000	n.s.
MATH	Qwen GPTQ-4	0.245	n.s.	1.000	n.s.
MATH	Qwen FP8	0.383	n.s.	1.000	n.s.
MATH	Llama GPTQ-4	0.619	n.s.	1.000	n.s.
MATH	Llama FP8	0.647	n.s.	1.000	n.s.
GSM8K	Llama AWQ-4	0.002	sig.	0.031	sig.
GSM8K	Llama GPTQ-4	0.511	n.s.	1.000	n.s.
GSM8K	Qwen AWQ-4	0.579	n.s.	1.000	n.s.
GSM8K	Qwen GPTQ-4	0.729	n.s.	1.000	n.s.
GSM8K	Llama FP8	0.761	n.s.	1.000	n.s.
GSM8K	Qwen FP8	0.827	n.s.	1.000	n.s.
GPQA	Qwen AWQ-4	0.008	border.	0.130	n.s.
GPQA	Qwen GPTQ-4	0.288	n.s.	1.000	n.s.
GPQA	Llama FP8	0.403	n.s.	1.000	n.s.
GPQA	Llama GPTQ-4	0.573	n.s.	1.000	n.s.
GPQA	Qwen FP8	0.690	n.s.	1.000	n.s.
GPQA	Llama AWQ-4	0.711	n.s.	1.000	n.s.

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